GGSS-R Extension

Overview

This directory contains a sequence of experiments investigating GGSS-R, the image reconstruction attack proposed in *Enhanced Privacy Leakage from Noise-Perturbed Gradients via Gradient-Guided Conditional Diffusion Models*. GGSS-R reconstructs a client's private training image from its leaked model gradient by combining two signals during reverse diffusion:

- **a diffusion prior**, which keeps the generated sample on the natural image manifold;
- **gradient guidance**, which steers the sample toward an image whose victim-model gradient matches the leaked gradient.

The paper evaluates reconstruction across several victim architectures and introduces Reconstruction Vulnerability (RV) to quantify how vulnerable different models are to gradient inversion. It also studies reconstruction from perturbed gradients and uses a fixed guidance rate m_r to balance gradient matching against the diffusion prior.

The experiments in this directory examine two aspects of that setup more closely.

Victim architecture is only part of reconstruction vulnerability.

The paper reports reconstruction vulnerability for different model architectures, but the released repository does not include the corresponding victim checkpoints or fully specify the trained state associated with those reconstruction results. Recreating the victim locally exposed an important additional factor: **the same architecture can be either highly reconstructable or effectively non-reconstructable depending on its training state and resulting gradient structure.**

A gradient can be nonzero and its distance can be successfully minimized without containing enough image-specific information to recover the input. Across the diagnostic experiments here, reconstruction failure is traced through the pipeline until the victim gradient itself is isolated. Layer-wise analyses then show that what matters is not simply model size, classification loss, or total gradient magnitude, but whether a sufficiently strong and informative gradient reaches the input-facing layers.

This extends the paper's architecture-focused analysis of reconstruction vulnerability by showing that gradient identifiability is also strongly dependent on the victim model's training state.

Gradient similarity is not necessarily image similarity.

GGSS-R guides reconstruction by reducing the distance between candidate and leaked gradients. Several experiments here test the assumption underlying that objective: whether moving closer in gradient space actually constrains the candidate toward the private image.

For poorly identifying victim states, it does not. Candidates can reach extremely high gradient cosine similarity while remaining far from the target in pixel and perceptual space. Population-level and layer-wise analyses are used to examine why this happens and which victim states instead produce gradients from which the image can genuinely be recovered.

Static guidance can overfit a perturbed gradient.

The paper also shows that under sufficiently noisy gradients, reconstruction quality can improve and then deteriorate during reverse diffusion. The authors attribute this behavior to gradient guidance eventually overpowering the diffusion prior and forcing the reconstruction toward the noise contained in the perturbed gradient.

The released method nevertheless uses a **static guidance rate** throughout reconstruction.

The final experiment extends this mechanism with a **linearly decaying guidance schedule**. Strong guidance is retained early to establish target correspondence, then reduced during later denoising so that the natural-image prior is less dominated by the noisy gradient. Under the tested Gaussian perturbation, this produces substantially better reconstruction than the paper's fixed guidance rate.

Experimental Progression

The experiments are intended to be read as a sequence. Each one isolates a question raised by the result immediately before it.

Baseline GGSS-R reconstruction

The sequence begins with a direct reconstruction attempt using the authors' released GGSS-R pipeline. Their MLP_1 victim is trained locally for one epoch, while a CelebA test image is used as the reconstruction target with the paper's fixed m_r = 0.2. The pipeline executes successfully, but the target is not recovered visually. This establishes the initial failure and motivates testing whether the victim and target setup is responsible.

More training and a training-member target

The next experiment trains the same victim for three epochs and replaces the held-out target with an explicitly class-0 image from the victim's training set. The expectation is that repeated exposure to the target may make its gradient more image-specific. Instead, reconstruction becomes substantially worse. More victim training and guaranteed training-set membership therefore do not solve the problem and suggest that the victim's learned state may itself be affecting reconstruction.

Unperturbed reproduction and pipeline diagnostics

The reconstruction is then returned to the single-epoch victim while replacing the generic diffusion checkpoint used earlier with the exact checkpoint referenced by the authors. The attack is kept unperturbed and instrumented with additional diagnostics. Reconstruction still fails, shifting the investigation away from simple target selection and checkpoint mismatch and toward isolating the individual components of GGSS-R.

Unconditional diffusion diagnosis

The gradient-guidance path is removed entirely and the referenced FFHQ diffusion model is tested on its own. Starting from Gaussian noise, it successfully produces natural face images. The diffusion prior is therefore functional and capable of representing the image domain required by the attack.

GSS without leaked-gradient guidance

The active GSS sampling machinery is then retained while the leaked-gradient guidance is disabled. The resulting samples remain on the natural face manifold even though, as expected, they do not reconstruct the private target. Together with the unconditional diffusion test, this makes a broken diffusion model or basic GSS sampling path unlikely to explain the reconstruction failures.

Direct gradient inversion without diffusion

Attention then shifts entirely to the leaked gradient. Diffusion is removed and the candidate image is optimized directly against the victim gradient. Gradient distance decreases substantially and cosine similarity can exceed 0.999, yet the reconstructed image remains far from the target. This establishes that the earlier failures are not solely caused by diffusion: under this victim state, matching the gradient itself is insufficient for image recovery.

Gradient identifiability analysis

The gradient is then studied directly across a balanced population of CelebA images and across individual victim layers. Gradient similarity is found to contain real image-related structure, but not enough to uniquely constrain the target under the tested single-epoch MLP_1. Most importantly, the input-facing layer carries very weak target-specific gradient signal compared with downstream classification layers. This provides a concrete explanation for why gradient optimization can succeed numerically while image reconstruction fails.

Victim architecture and training-state comparison

The victim is next treated as the experimental variable. Four custom architectures are compared at initialization and after one epoch of training using direct inversion of clean gradients. The untrained 25.2M-parameter MLP reconstructs the target consistently across all seeds, reaching a mean MSE of **0.00113** and PSNR of **29.72 dB**. After one epoch, the same architecture becomes much harder to reconstruct. All three CNN architectures fail even at initialization: their total gradients may be large, but the gradient signal largely vanishes before reaching their early feature layers. This demonstrates that reconstruction vulnerability depends on **both architecture and model state**, specifically on whether image-identifying gradients survive to input-connected layers.

Scheduled guidance for perturbed gradients

With a victim state whose clean gradient has been shown to support reconstruction, the final experiment returns to GGSS-R's intended perturbed-gradient setting. At Gaussian noise variance 0.01, the paper's fixed m_r = 0.2 is compared with a linear schedule that decays from 0.2 to 0 throughout reverse diffusion. The scheduled run increasingly outperforms static guidance as reconstruction progresses, finishing at **0.0548 MSE, 12.61 dB PSNR, and 0.8744 LPIPS**, compared with **0.1087 MSE, 9.64 dB PSNR, and 0.9317 LPIPS** for fixed guidance. The result shows that reducing gradient influence during later denoising can better preserve reconstruction quality when the leaked gradient itself is noisy.

Overall Summary

The experiments separate two questions that can otherwise become conflated in gradient-guided reconstruction:

1. Does the victim gradient contain enough image-specific information to reconstruct the input?
2. If it does, how should that gradient be used when it has been perturbed?

The diagnostic sequence shows that the first question depends strongly on the victim's architecture, training state, and layer-wise gradient propagation. Successful optimization of a gradient-matching objective is not by itself evidence that the private image is recoverable.

Once a demonstrably identifiable victim gradient is established, the perturbed-gradient experiment addresses the second question. There, replacing GGSS-R's fixed guidance strength with a decreasing schedule improves the balance between target guidance and the diffusion prior.

Together, these experiments extend the GGSS-R evaluation in two directions: characterizing reconstruction vulnerability through gradient identifiability rather than architecture alone, and introducing scheduled guidance as an alternative to static guidance for reconstruction from perturbed gradients.

GGSS-R Baseline: Single-Epoch Victim and Public Diffusion

1. Motivation & Objective

This is the first reconstruction experiment in the diagnostic sequence. Its purpose is to establish a baseline reproduction of the GGSS-R image reconstruction pipeline using the authors' released implementation.

The authors' method reconstructs an image from a victim model's leaked gradient by combining a diffusion prior with gradient-guided sampling. Before investigating why the method may fail, we first need to reproduce the complete pipeline as closely as possible and determine whether the reconstruction works under a reasonable implementation of the authors' setup.

There are two important limitations in the released materials. The authors' repository does **not** provide the trained victim-model checkpoint used in their experiments, and the required FFHQ diffusion checkpoint is also **not** bundled with the code. We therefore construct the missing parts ourselves while keeping the attack implementation as close as possible to the authors' release.

For this baseline, the victim model is trained for **one epoch** on CelebA using the `Smiling` attribute. The reconstruction target is taken from the **CelebA test split**, meaning that the victim has not seen this image during training. The target is also selected directly by dataset index without first checking whether its class agrees with the class assumption in the authors' reconstruction code.

This setup intentionally gives us a direct starting point rather than modifying the experiment in advance to favor reconstruction. If the reconstruction fails, the result gives us a baseline from which specific possible causes can be investigated in subsequent experiments.

2. Methodology

The experiment begins by obtaining the authors' GGSS-R implementation. It clones the authors' public repository and extracts its `code.zip` bundle. The expected runner, diffusion implementation, measurement operator, victim model, and configuration files are checked before proceeding.

The victim model follows the authors' `MLP_1` architecture. It is a three-layer fully connected network operating on 256 × 256 RGB images:

$3 \times 256 \times 256 \rightarrow 512 \rightarrow 128 \rightarrow 2$

The model contains approximately 100 million parameters. It is trained on the CelebA training split for one epoch using AdamW with batch size 64, learning rate 10^{-4}, and weight decay 10^{-5}.

The resulting checkpoint is then evaluated on the held-out CelebA test split to verify that the victim model is functional before it is used for reconstruction.

Images follow the preprocessing expected by the authors' implementation:

Resize(256 × 256)
-- ToTensor()
-- Normalize(mean=0.5, std=0.5)

The reconstruction target is `CelebA test[0]`. Its original image and metadata are saved as experiment artifacts, while the authors' runner performs the corresponding resize and normalization when loading the target.

For the diffusion prior, the experiment uses `ffhq_10m.pt`, the FFHQ diffusion checkpoint expected by the authors' code. Because the checkpoint is not included in the repository, it is downloaded from a public mirror and verified using its expected SHA-256 hash before use.

The attack itself is then executed through the authors' `sample_condition_same_inputs.py` runner using:

- GSS conditioning;
- the authors' 1000-step DDIM configuration;
- guidance rate $m_r = 0.20$;
- batch size 1 for gradient acquisition;
- no additional gradient perturbation, i.e. the runner uses the target gradient directly.

Only two small changes are made to the authors' source code. The reconstruction progress directory is created before the sampler writes to it, and the diffusion trajectory is saved sparsely rather than at every timestep. These changes affect output handling only; the GSS calculation, gradient guidance, diffusion equations, and reconstruction logic are left unchanged.

The reconstruction is evaluated using both the saved noisy diffusion states x_t and the predicted clean-image states x_0. MSE, PSNR, and LPIPS are calculated against the target image at the saved timesteps.

$$MSE = \frac{1}{H \cdot W \cdot C} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{c=1}^{C} (I_{i,j,c} - K_{i,j,c})^2$$

where,

- **H, W, C:** Image spatial dimensions—Height, Width, and number of color Channels (e.g., C = 3 for RGB images).
- **I_(i,j,c):** Pixel value of the ground-truth target image at spatial coordinate (i, j) and color channel c.
- **K_(i,j,c):** Pixel value of the reconstructed candidate image at spatial coordinate (i, j) and color channel c.

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

where,

- **MAX_I:** Maximum dynamic range of pixel values (1.0 for normalized tensors in [-1, 1] or [0, 1]; 255 for standard 8-bit integer images).
- **MSE:** Mean Squared Error calculated between the target image I and candidate image K.
- **log_10:** Base-10 logarithm, measuring image fidelity on a logarithmic decibel (dB) scale.

$$LPIPS = d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} ||w_l \odot (\hat{y}_{l,h,w} - \hat{y}_{0,l,h,w})||_2^2$$

This allows the experiment to evaluate not only the final reconstruction but also how image similarity changes throughout the reverse-diffusion trajectory.

3. Results & Observations

The baseline victim model trained successfully and reached 86.97% training accuracy and 86.84% held-out test accuracy after one epoch. The target was successfully loaded and processed, and the full 1000-step GGSS-R reconstruction completed successfully using the authors' code path.

The reconstruction itself, however, did not recover the target image.

The quantitative trajectory shows a particularly important pattern. The final predicted x_0 at timestep 0 has:

- MSE: **0.006668**
- PSNR: **21.76 dB**
- LPIPS: **0.4560**

These values are substantially better than the later three-epoch experiment, but they do not by themselves establish successful reconstruction. They are far from the paper's reported reconstruction quality of MSE = 4.55e-5, PSNR = 43.42, and LPIPS = 1.60e-7. The MSE is about 150 times larger than what is expected, while the PSNR is two times smaller.

timestep	x0_mse	x0_psnr	x0_lpips
0	0.006668379064649343 5	21.75979720630726	0.45600005984306335
100	0.007661618292331695 6	21.156794886963837	0.486350417137146
200	0.00951185636222362 5	20.217347165858193	0.5227426886558533
300	0.01394592225551605 2	18.555527601905077	0.6198112368583679
400	0.02652912959456443 8	15.762769987353035	0.7571112513542175
500	0.04259172827005386 4	13.706747369415343	0.8655551075935364
600	0.06281916797161102	12.019078200503934	1.0273545980453491
700	0.09846444427967072	10.06720565855794	1.1581746339797974
800	0.13785669207572937	8.605721467796908	1.3718396425247192
900	0.1668679416179657	7.776270912161442	1.4098410606384277

Table: Saved trajectory of pixel-space metrics

The saved trajectory of the pixel-space metrics for the reconstructed images shows that reconstruction quality is improving steadily across all three metrics. The MSE and LPIPS are consistently decreasing, while PSNR is consistently increasing. The best predicted-x_0 image under all three reported metrics thus occurs at the final saved timestep t = 0, after a 1000 reconstruction steps have completed. The problem is that, despite the improvements, the final reconstructed image is still nowhere close to the target-image in pixel-space.

Figure: Visual reconstruction grid recorded at every 100 timesteps for single-epoch victim with arbitrary target image.

The visual reconstruction also does not provide a clean recovery of the target. No natural facial features can be discerned visually, even from the last reconstructed image. The attack therefore does not reproduce the strong reconstruction behavior reported by the authors under this independently constructed victim setup.

At this stage, however, the failure has several possible explanations. The victim was trained for only one epoch, the target was a held-out image rather than a training member, and the target was selected without explicitly checking the class assumption embedded in the authors' released reconstruction code. In addition, the diffusion checkpoint used here was obtained from a public mirror rather than being supplied directly with the GGSS-R repository.

The next experiment therefore changes the two most obvious victim/target conditions: the victim is trained for longer, and the target is explicitly selected as a class-0 member of the victim's training set. The purpose is to test whether a more strongly fitted victim and a training-member target produce a more image-specific leaked gradient and improve reconstruction.

GGSS-R Training-Member Reconstruction with a More-Trained Victim

1. Motivation & Objective

This experiment follows the baseline GGSS-R reproduction and tests whether the victim model and target selection used there were responsible for the poor reconstruction.

In the baseline experiment, the victim was trained for only one epoch and the reconstruction target was taken from the CelebA test split. The target therefore had not been seen by the victim during training. The target was also selected without checking its `Smiling` label, even though the authors' released reconstruction operator contains a class-0 assumption.

This experiment changes both conditions.

First, the victim is trained for **three epochs instead of one**. The motivation is that a more strongly fitted victim may memorize its training examples more effectively. If the leaked gradient becomes more image-specific as the victim fits the training data more closely, reconstruction could become easier.

Second, the target is explicitly selected from the **training split** and required to have `Smiling == 0`. This makes the target a genuine member of the victim's training data while also satisfying the class convention assumed by the authors' released reconstruction code.

The experiment therefore tests the hypothesis that:

A more strongly trained victim combined with a training-member target may produce a more image-specific leaked gradient and improve GGSS-R reconstruction.

The GGSS-R attack itself is not redesigned. The goal is to determine whether changing the victim-training and target-membership conditions is enough to recover the target.

2. Methodology

The experiment uses the same authors' `MLP_1` victim architecture and GGSS-R reconstruction pipeline as the baseline.

The victim is trained on the complete CelebA training split using the same preprocessing and optimization settings as before:

- batch size: 64;
- AdamW learning rate: 10^{-4};

- weight decay: 10^{-5};
- training duration: 3 epochs.

A checkpoint is saved after every completed epoch, with the third-epoch checkpoint used as the canonical attack checkpoint. The victim contains 100,729,730 parameters.

The target is selected deterministically as the first CelebA training example with `Smiling == 0`. This image is therefore both:

1. a member of the data used to train the victim, and
2. consistent with the class-0 assumption in the authors' ReconstructionOperator.

The target is saved once as an experiment artifact and reused throughout the reconstruction.

The same FFHQ diffusion checkpoint used in the baseline experiment is used here. It is downloaded and SHA-256 verified before being placed at the path expected by the authors' code.

The reconstruction again uses the authors' GSS implementation with:
m_r = 0.20, 1000 DDIM diffusion steps, and no additional gradient noise.

The notebook makes only three compatibility/output changes:

- represents the hard-coded class-0 target as a `LongTensor` class index;
- creates the reconstruction progress directory before it is used;
- saves sparse x_t and predicted x_0 snapshots every 100 steps and at t = 0.

The underlying GSS calculation and GGSS-R sampling procedure remain unchanged.

After reconstruction, the saved trajectory is evaluated using MSE, PSNR, and LPIPS against the target. Both the noisy x_t states and the predicted clean x_0 states are evaluated so that the reconstruction can be inspected numerically and visually across the full diffusion process.

3. Results & Observations

The three-epoch victim trained successfully. Its training loss decreased from **0.3066 after epoch 1 to 0.2384 after epoch 3**, and the final held-out test accuracy was **89.97%**. The target was confirmed to be a class-0 training member, and the trained victim also classified it as class 0.

Thus, the two changes motivated by the baseline experiment were successfully implemented: the victim was more strongly trained, and the target was appropriately selected.

They did **not** improve the reconstruction.

The full 1000-step GGSS-R run completed, but the reconstruction trajectory remained poor. At the final predicted x_0, the image-space metrics were:

- MSE: **0.172187**
- PSNR: **7.64 dB**
- LPIPS: **0.718507**

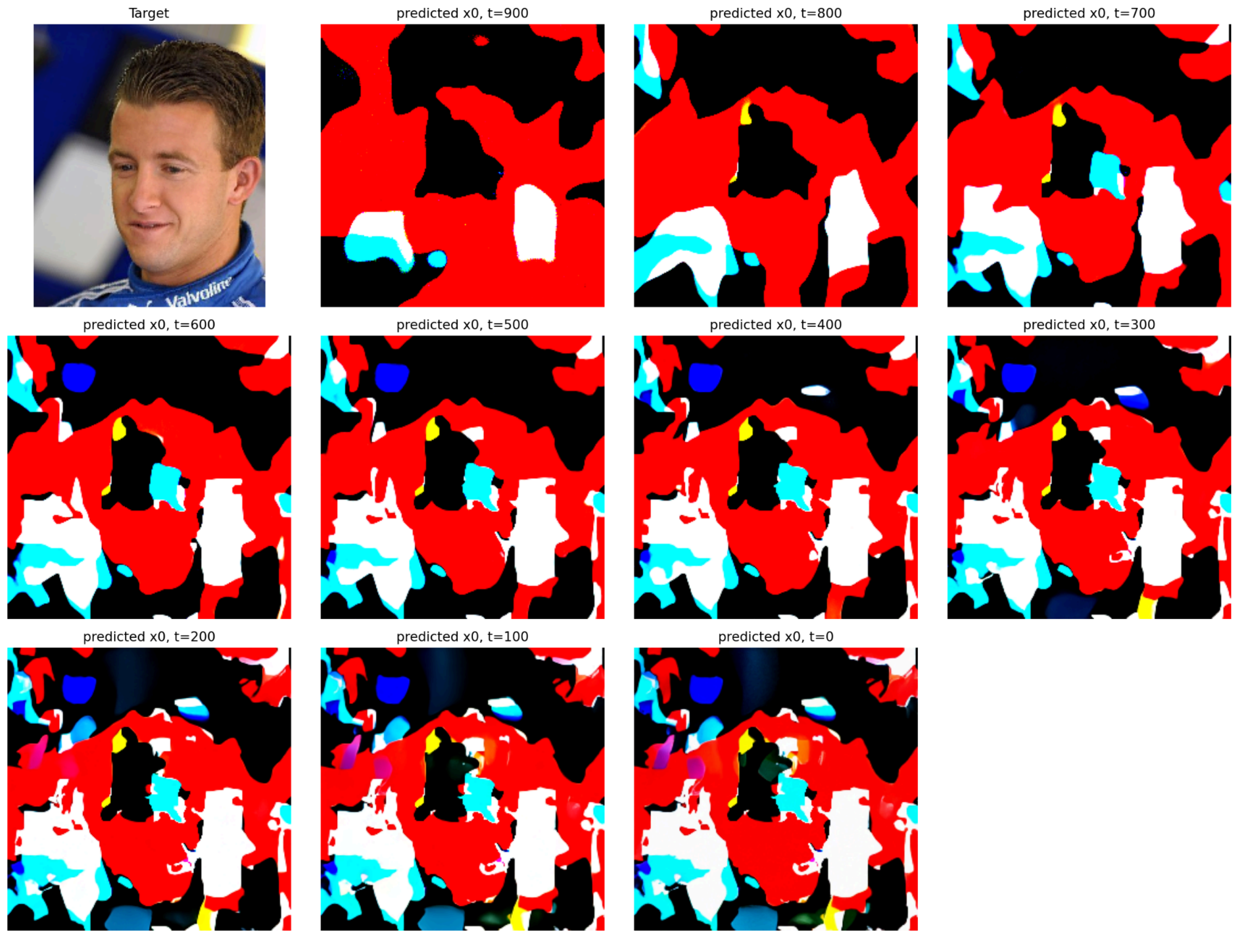

Figure: Visual reconstruction grid recorded at every 100 timesteps for three-epoch-trained victim and class-0 target image from train set.

The visual reconstruction is even more revealing: the final outputs do not recover meaningful features of the target face and instead become dominated by patches of color and distorted structures.

Therefore, the hypothesis tested here is not supported. Making the victim substantially more trained and choosing a genuine training-member target did not rescue GGSS-R reconstruction; in this run, the result was substantially worse than the first baseline.

This changes the direction of the investigation. The failure is no longer well explained simply by the victim being insufficiently trained or the target being outside the training set. We now need to examine the reconstruction pipeline itself more carefully.

The next experiment, `ggssr_unperturbed_reproduction_diagnostic`, therefore moves away from changing the victim and instead performs a more controlled reproduction of the GGSS-R setup. In particular, it uses the FFHQ checkpoint referenced by the DPS implementation and adds a series of pre-reconstruction diagnostics to test the target gradient, the gradient of unrelated images, second-order differentiability, the gradient behavior of an actual diffusion sample, and finally the authors' original ReconstructionOperator. These checks are intended to determine whether the reconstruction failure can be localized to a specific component of the pipeline.

GGSS-R Unperturbed Reconstruction Diagnostic

1. Motivation & Objective

This experiment is the first major diagnostic step after the two initial reconstruction attempts.

The baseline experiment used a one-epoch victim and a held-out target, while the second experiment used a three-epoch victim and a training-member target. Neither produced the expected reconstruction. The second experiment in particular showed that simply training the victim longer and giving the attack a target that it had actually seen during training did not solve the problem.

At this point, changing the victim-training conditions further would not tell us whether the problem lies elsewhere in the reconstruction pipeline. We therefore return to a controlled GGSS-R reproduction and inspect the individual components before and after the actual reconstruction.

A further issue in the baseline setup also needs to be addressed. The authors' GGSS-R repository does not itself contain the required diffusion checkpoint. This experiment therefore uses the **FFHQ diffusion checkpoint referenced by the paper**, rather than treating an arbitrary public diffusion checkpoint as interchangeable with the one expected by the authors' pipeline.

The main objective is to answer a more basic question:

Is the GGSS-R reconstruction pipeline actually receiving valid gradients, computing the gradient-distance objective correctly, and propagating that objective back to the diffusion image as intended?

The experiment does not assume that the final reconstruction must succeed. Instead, it establishes whether the relevant pieces of the pipeline are functioning before interpreting the eventual reconstruction failure.

This is important for the later diagnostic sequence. If the diffusion model itself is not generating valid images, that needs to be isolated. If the victim gradient is invalid, that needs to be isolated. If the gradient-distance objective cannot be differentiated with respect to the image, the GGSS-R guidance mechanism cannot work. And if all of these checks pass but reconstruction still fails, attention can move toward the information contained in the leaked gradient itself.

2. Methodology

The experiment is built around the authors' released GGSS-R implementation rather than reimplementing the attack from scratch.

The public repository is cloned and its code.zip bundle is extracted. The notebook records the upstream commit and verifies that the expected runner, diffusion implementation, victim model, measurement operator, and configuration files are present.

The FFHQ diffusion checkpoint ffhq_10m.pt is obtained from the public checkpoint location referenced by the DPS project. The checkpoint is loaded and verified before being placed at the path expected by GGSS-R.

The victim follows the authors' MLP_1 architecture and is trained using the established experiment procedure. A deterministic class-0 CelebA training-member target is constructed and stored as an immutable target artifact. The target and victim checkpoint are then verified together before reconstruction.

Three small compatibility/output patches are applied to the authors' source:

1. create the reconstruction progress directory before it is written to;
2. create the predicted-x_0 directory before those snapshots are written;
3. save sparse trajectory snapshots rather than writing an image at every one of the 1000 diffusion steps.

These patches change file handling only and are not intended to modify the GGSS-R reconstruction equations.

Before running the attack, the notebook performs five diagnostics.

Target-gradient validity

The target image is passed through the victim model and its full parameter gradient is computed. The diagnostic checks the gradient norm, the number of nonzero elements, and whether the autograd graph is preserved.

This establishes that the target produces a valid measurement for GGSS-R.

Unrelated-image gradient validity

A different CelebA image is passed through the same victim and loss function. Its gradient is compared at the basic numerical level with the target gradient.

The purpose is to make sure the target is not an unusual case where the victim-gradient mechanism itself is malfunctioning. A normal unrelated image should also produce a valid, nonzero gradient.

Second-order differentiability

GGSS-R does not only require the victim gradient $g(x) = L(f(x), y)$.

It needs to differentiate a distance between the target gradient and the candidate gradient back through the candidate image. Conceptually, the required path is

$x \rightarrow g(x) \rightarrow ||g(x) - g\_target|| \rightarrow \nabla\_x \, ||g(x) - g\_target||$.

The notebook explicitly constructs this path and checks that the resulting image gradient is finite and nonzero.

Gradient behavior on an actual diffusion sample

The previous checks use real CelebA images. GGSS-R, however, applies the gradient guidance to images produced by the FFHQ diffusion process.

The notebook therefore generates an unconditional 1000-step DDIM sample using the same FFHQ diffusion model and checks whether this generated image also produces a valid victim gradient and a differentiable gradient-distance objective.

This isolates the interaction between the diffusion image representation and the victim-gradient measurement before the full guided reconstruction is attempted.

Original GGSS-R measurement operator

Finally, the same diffusion sample is passed through the authors' original `ReconstructionOperator.forward()` implementation.

Its gradient output and target-to-sample gradient distance are compared against the values obtained from the controlled diagnostic implementation. Exact agreement is used as a consistency check that the authors' measurement operator is computing the expected quantity.

After these checks, the notebook executes the original GGSS-R reconstruction using GSS conditioning, 1000 DDIM steps, and

$m_r = 0.20$,

with no additional gradient perturbation.

The reconstruction trajectory is then analyzed using MSE, PSNR, LPIPS, victim-gradient distance, relative gradient distance, and gradient cosine similarity.

$$D_{\text{abs}}(g(x), g_{\text{target}}) = \|g(x) - g_{\text{target}}\|_2$$

where,

g(x): Gradient of the victim model's loss computed with respect to the candidate image x.

g_target: Leaked ground-truth gradient computed from the target image x_target.

|| · ||_2: L2 norm (Euclidean distance), measuring the absolute magnitude of the gradient error vector across all model parameters.

$$D_{\text{rel}}(g(x), g_{\text{target}}) = \frac{\|g(x) - g_{\text{target}}\|_2}{\|g_{\text{target}}\|_2}$$

where,
g(x): Candidate gradient evaluated at image x.
g_target: Target gradient leaked from the victim model.
|| g(x) - g_target ||_2: L2 norm of the absolute difference between the candidate and target gradients.
|| g_target ||_2: L2 norm of the target gradient, used as a normalizing factor so that the distance is invariant to overall gradient scale.

$$S_{\cos}(g(x), g_{\text{target}}) = \frac{g(x) \cdot g_{\text{target}}}{\|g(x)\|_2 \; \|g_{\text{target}}\|_2}$$

where,
g(x) · g_target: Dot product (inner product) between the candidate and target gradient vectors.
|| g(x) ||_2, || g_target ||_2: L2 norms of the candidate and target gradients, respectively.
S_cos: Cosine of the angle between the two gradient vectors, bounded in [-1, 1], where 1 indicates perfect directional alignment regardless of vector magnitude.

The post-reconstruction analysis also decomposes the remaining gradient mismatch by victim-model layer.

3. Results & Observations

The pre-reconstruction diagnostics consistently passed.

The target produced a nonzero full parameter gradient with norm approximately 1.3358.

There were 17,107,415 nonzero gradient elements out of 100,729,730, and the autograd graph was preserved.

The unrelated CelebA image also produced a valid gradient, with norm approximately 35.28 and more than 22.6 million nonzero elements. This shows that the victim-gradient measurement is not failing only for the target.

The second-order differentiability test also passed. The gradient-distance objective had value approximately 35.86, and its gradient with respect to the image had norm approximately 11.68, with all 196,608 input pixels receiving a nonzero gradient. This is important because it demonstrates that the gradient-matching objective can actually propagate information back to the candidate image.

The diffusion-specific diagnostic also passed. The DPS FFHQ model successfully generated a 256×256 sample through the full 1000-step DDIM process. That generated image produced a valid victim gradient with norm approximately 2.839, and its gradient differed from the target gradient by approximately 2.801. The gradient-distance objective also propagated a nonzero gradient back to the diffusion image.

Lastly, the authors' original ReconstructionOperator reproduced the same gradient-distance result: 2.8006341457 with an absolute difference of 0 from the controlled diagnostic calculation.

These checks substantially narrow down the possible sources of failure. The target gradient is valid, unrelated images produce valid gradients, the gradient-distance objective is differentiable with respect to image pixels, the diffusion model produces usable samples, and the authors' original measurement operator agrees with the controlled implementation.

The full unperturbed GGSS-R reconstruction nevertheless fails to recover the target.

The 1000-step run completes, but the reconstruction trajectory does not approach the target in image space. The final reconstruction has:

- pixel MSE: **0.173091**
- PSNR: **7.617 dB**
- LPIPS: **0.688771**
- victim-gradient distance: **0.428106**
- relative gradient distance: **0.320480**
- gradient cosine similarity: **0.948341**

The reconstruction therefore does achieve a reasonably strong alignment with the target gradient, but this does not translate into a visually accurate reconstruction. The saved trajectory shows the same separation: by the final step, the gradient cosine similarity is high while the image remains substantially different from the target.

The comparison with three unrelated class-0 CelebA images is also informative. Those images have pixel MSEs around **0.23 - 0.24** from the target, while their relative gradient distances are much larger, approximately **23 - 87**. The GGSS-R reconstruction is considerably closer to the target in gradient space than these comparison images, but it is still far from the target in image space.

Parameter	Squared Gradient Distance	Contribution Percent
fc2.weight	0.13743	74.98546%
fc1.weight	0.04007	21.86483%
fc3.weight	0.00577	3.14830%
fc3.bias	0.00000	0.00076%
fc2.bias	0.00000	0.00059%
fc1.bias	0.00000	0.00006%

Table: Layer-wise decomposition of contribution to gradient distance.

The layer-wise decomposition shows that the remaining target-to-reconstruction gradient mismatch is concentrated mainly in:

- fc2.weight: **74.99%**
- fc1.weight: **21.86%**
- fc3.weight: **3.15%**

with the bias terms contributing negligibly.

Taken together, these results change the direction of the investigation. The reconstruction failure is not readily explained by an invalid target gradient, a broken second-order gradient path, an unusable diffusion sample, or a mismatch between the controlled measurement and the authors' ReconstructionOperator.

The remaining question is therefore not simply whether GGSS-R is capable of minimizing gradient distance. It clearly can. The more important question is whether minimizing the leaked-gradient distance actually provides enough information to identify the target image.

This motivates the next diagnostics. The diffusion model and the GSS execution path are first tested independently in `unconditional_diffusion_manifold_diagnosis` and `gss_with_no_guidance_diagnosis`. Direct gradient inversion is then tested without diffusion in `direct_gradient_inversion_wo_diffusion`. Finally, `gradient_identifiability_analyses` examines the relationship between image space and gradient space more systematically to determine whether the leaked gradient from the MLP victim is sufficiently image-specific for reconstruction.

Unconditional Diffusion Manifold Diagnosis

1. Motivation & Objective

The GGSS-R reconstruction pipeline relies on the authors' FFHQ diffusion model as its image prior. When the full reconstruction produces poor images, one possible explanation is that the diffusion model itself is not generating meaningful images in the first place.

This experiment isolates that component completely.

The diffusion model is used **without** a victim model, target image, leaked gradient, measurement, GSS conditioning, or reconstruction objective. Starting only from Gaussian noise, the experiment asks a simple question:

Can the authors' FFHQ diffusion model, using the original implementation and checkpoint, independently denoise random noise into natural-looking human-face images?

This is an important control experiment. If unconditional sampling works, then the poor reconstruction observed in the earlier GGSS-R experiments cannot simply be attributed to an incapable diffusion prior. The investigation can instead move to the interaction between the diffusion model and the leaked-gradient guidance.

The experiment also establishes a clean reference for the next diagnostic, `gss_with_no_guidance_diagnosis.ipynb`. There, the same diffusion machinery is placed back inside the authors' GSS reconstruction path, but the gradient-guidance coefficient is set to zero.

2. Methodology

The experiment uses a fresh, isolated workspace so that the diffusion diagnostic does not modify the earlier reconstruction experiments.

Authors' implementation and checkpoint

The authors' GGSS-R repository is cloned, the packaged `code.zip` is extracted and the required diffusion files are verified before the model is loaded.

The FFHQ diffusion checkpoint `ffhq_10m.pt` is copied into the experiment directory and verified using SHA-256.

The checkpoint is 357.07 MB and contains 362 parameter tensors. The loaded U-Net is also compared against the checkpoint parameter-by-parameter; all 362 tensors match with a maximum absolute difference of 0.0.

Diffusion configuration

The experiment keeps the authors' diffusion configuration rather than introducing a separate sampler implementation.

The main settings are:

- Image size: 256 × 256
- Model: authors' FFHQ U-Net
- Sampler: DDIM
- Diffusion steps: 1000
- Noise schedule: linear

The loaded U-Net contains 93,563,910 (93.5M) parameters.

Unconditional sampling

Four independent trajectories are generated.

For each sample, the initial state is pure Gaussian noise:

$x_{999} \sim N(0, I)$.

The experiment then traverses the authors' DDIM trajectory backwards:

$x_{999} \rightarrow x_{998} \rightarrow \ldots \rightarrow x_0$.

At every step, the authors' own `sampler.p_sample()` implementation is called.

No conditioning function is invoked:

- no victim model,
- no target image,
- no gradient,
- no measurement,
- no GSS,
- no gradient guidance.

Selected x□ states and `pred_xstart` estimates are saved every 100 steps so that the complete denoising trajectory can be inspected without storing every intermediate image.

For visualization, the primary conversion is the fixed model-range mapping

x_image = (x + 1) / 2.

This converts the diffusion-generated images with pixel range [-1, 1] to a displayable [0, 1] format. The authors' `clear_color()` min-max normalization is not used as the primary representation because independently rescaling each image can make poor absolute scaling appear visually better than it actually is.

Finally, the four generated samples are collected into a contact sheet and an experiment manifest records the implementation, checkpoint, configuration, seed, device, and runtime.

3. Results & Observations

All four unconditional diffusion trajectories completed successfully. Each required the full 1000-step DDIM trajectory and took approximately 1.7 minutes on the Tesla T4 used for the experiment.

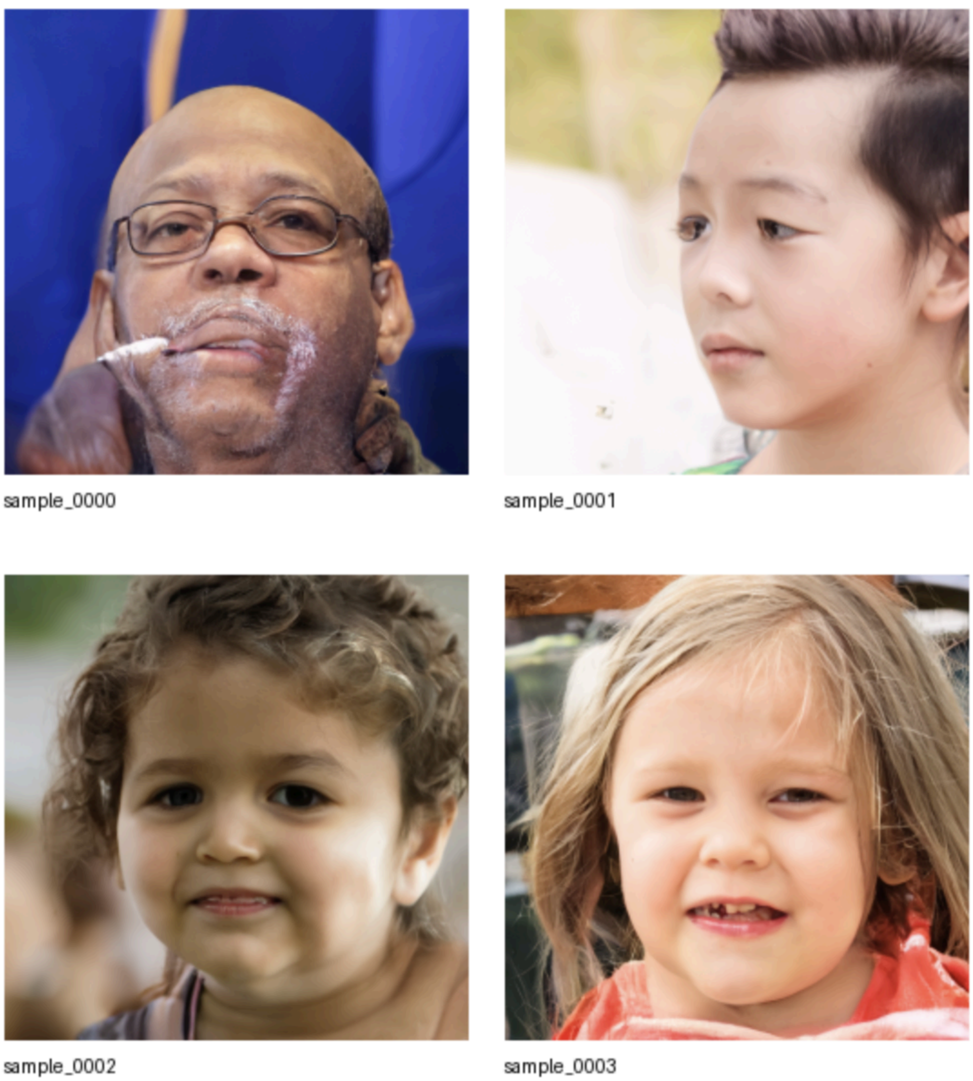

Figure: Final denoised images generated by unconditional DDIM diffusion sampling.

The generated samples are natural-looking human-face images rather than unresolved noise. The four outputs also show that the behavior is not limited to a single successful random seed: all four independent trajectories reach coherent face images.

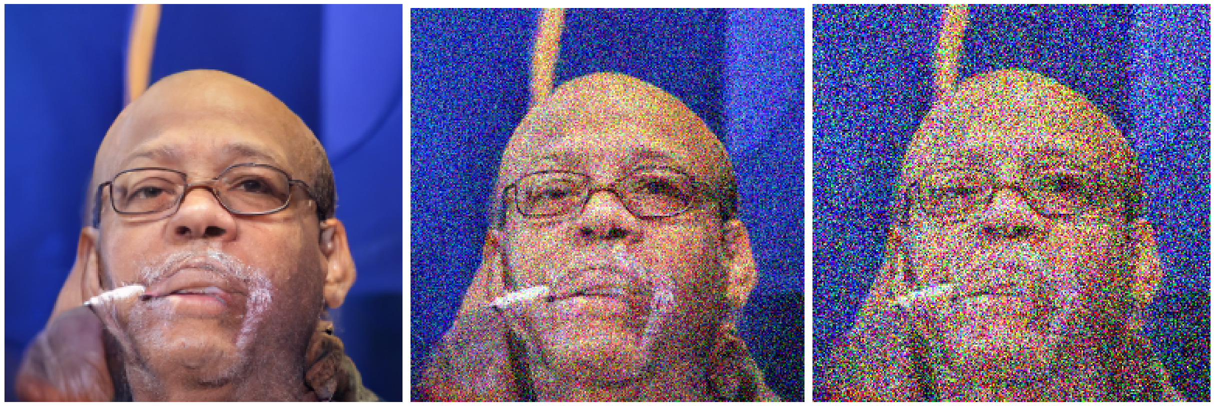

Figure: Trajectory of samples generated at the 1000th, 900th and 800th steps respectively.

The trajectory inspection provides the same conclusion from another perspective. Starting from Gaussian noise, the diffusion process progressively removes noise and produces structured human-face content. The final contact sheet shows all four independent samples together.

Therefore, this diagnostic supports the diffusion prior as a working component of the pipeline.

This does not prove that the diffusion prior will reconstruct the correct target when conditioned on a leaked gradient. It only establishes that, when used by itself, the model can successfully move random noise onto a meaningful human-face image manifold.

That distinction is important for the next step. The full GGSS-R failure cannot be explained simply by saying that the diffusion model is incapable of producing faces. The next diagnostic therefore keeps the same diffusion model but places it back inside the authors' GSS reconstruction pipeline while setting the gradient-guidance coefficient to

$m_r = 0$.

If that GSS-only path also produces meaningful face images, the remaining question becomes much more focused. That is, is the problem caused by the leaked gradient not providing enough image-specific information for successful reconstruction?

GSS with Zero Gradient Guidance Diagnosis

1. Motivation & Objective

The previous unconditional_diffusion_manifold_diagnosis.ipynb showed that the authors' FFHQ diffusion model can independently denoise Gaussian noise into natural-looking human-face images.

However, that experiment does not test the actual GGSS-R reconstruction path. It bypasses the GSS conditioning mechanism completely.

The earlier full GGSS-R reproduction, on the other hand, uses gradient guidance and produced poor reconstruction results. This leaves an important implementation question:

Is the reconstruction failure caused by the diffusion/GSS sampling path itself, or specifically by the gradient-guidance component?

This experiment isolates that question by running the authors' **GSS reconstruction pipeline with gradient guidance disabled**:

$m_r = 0$.

Everything else is kept fixed as closely as possible:

- authors' repository and configuration,
- FFHQ diffusion checkpoint,
- victim-model checkpoint,
- CelebA target image,
- reconstruction measurement,
- GSS conditioning mechanism,
- 1000-step DDIM sampling.

The intended algorithmic change is only the removal of gradient guidance.

This creates a useful second control point.

If GSS with $m_r = 0$ still produces meaningful face images, then the diffusion prior and the basic GSS sampling path are both capable of producing valid images. The poor result at $m_r > 0$ would then become much less likely to be caused by a basic diffusion or GSS implementation failure.

The next investigation can consequently focus on the remaining explanation: whether the leaked victim gradient contains enough information to identify the target image.

2. Methodology

Isolated experiment environment

A completely separate workspace is created with dedicated directories for:

- the authors' repository,
- copied input artifacts,
- reconstruction outputs,
- experiment logs.

The inputs from the earlier unperturbed reproduction are treated as read-only sources. The FFHQ checkpoint, victim checkpoint, and target image are copied into the new workspace before execution.

This separation makes the ablation easier to reproduce and ensures that the earlier experiment is not modified.

Authors' code and fixed inputs

A fresh checkout of the authors' repository is created, the `code.zip` folder is extracted and the required reconstruction files are checked, including:

- `sample_condition_same_inputs.py`
- `guided_diffusion/condition_methods.py`
- `guided_diffusion/measurements.py`
- `guided_diffusion/gaussian_diffusion.py`
- `guided_diffusion/attacked_model.py`
- the model and diffusion configuration files.

The FFHQ checkpoint is copied from the earlier reproduction and verified against its known SHA-256.

Output-only compatibility patches

Two small patches are applied to the authors' runner.

The first creates the reconstruction progress directory before it is used.

The second reduces intermediate image saving so that selected states are stored every 100 diffusion steps rather than writing images at every step.

These changes affect only output handling. They do not change the GSS calculation or the diffusion update itself.

Zero-guidance reconstruction

The authors' original reconstruction entry point is then executed with:
--method GSS
--guidance_scale 0.0

The GSS machinery remains active, but the gradient-guidance coefficient is zero: $m_r = 0$.

Conceptually, the comparison is therefore:
Full GGSS-R: $m_r > 0$ versus GSS-only ablation: $m_r = 0$.

The reconstruction still runs through the complete 1000-step diffusion trajectory. The experiment records the distance reported by the authors' code and saves sparse x_t and x_0 states for visual inspection.

This is different from the unconditional diffusion experiment. The unconditional test directly calls the DDIM sampler without any GSS machinery, whereas this experiment deliberately keeps the GSS reconstruction path active and removes only its gradient-guidance contribution.

3. Results & Observations

The zero-guidance reconstruction completed successfully for all 1000 diffusion steps.

The run used: guidance_scale = 0.0
and took approximately 31.41 minutes. The final reported distance was approximately *60.8*.

The distance does not decrease monotonically. It fluctuates substantially throughout the trajectory and ends around 60.8. This is expected, as the diffusion sampler is not receiving the leaked gradient guidance and thus has no incentive of minimizing the gradient distance. The guidance rate is intentionally set to 0, while keeping the GSS path active. This isolates differences in observation to the implementation of GSS, without being confounded by a leaked gradient.

The more important result comes from the generated images.

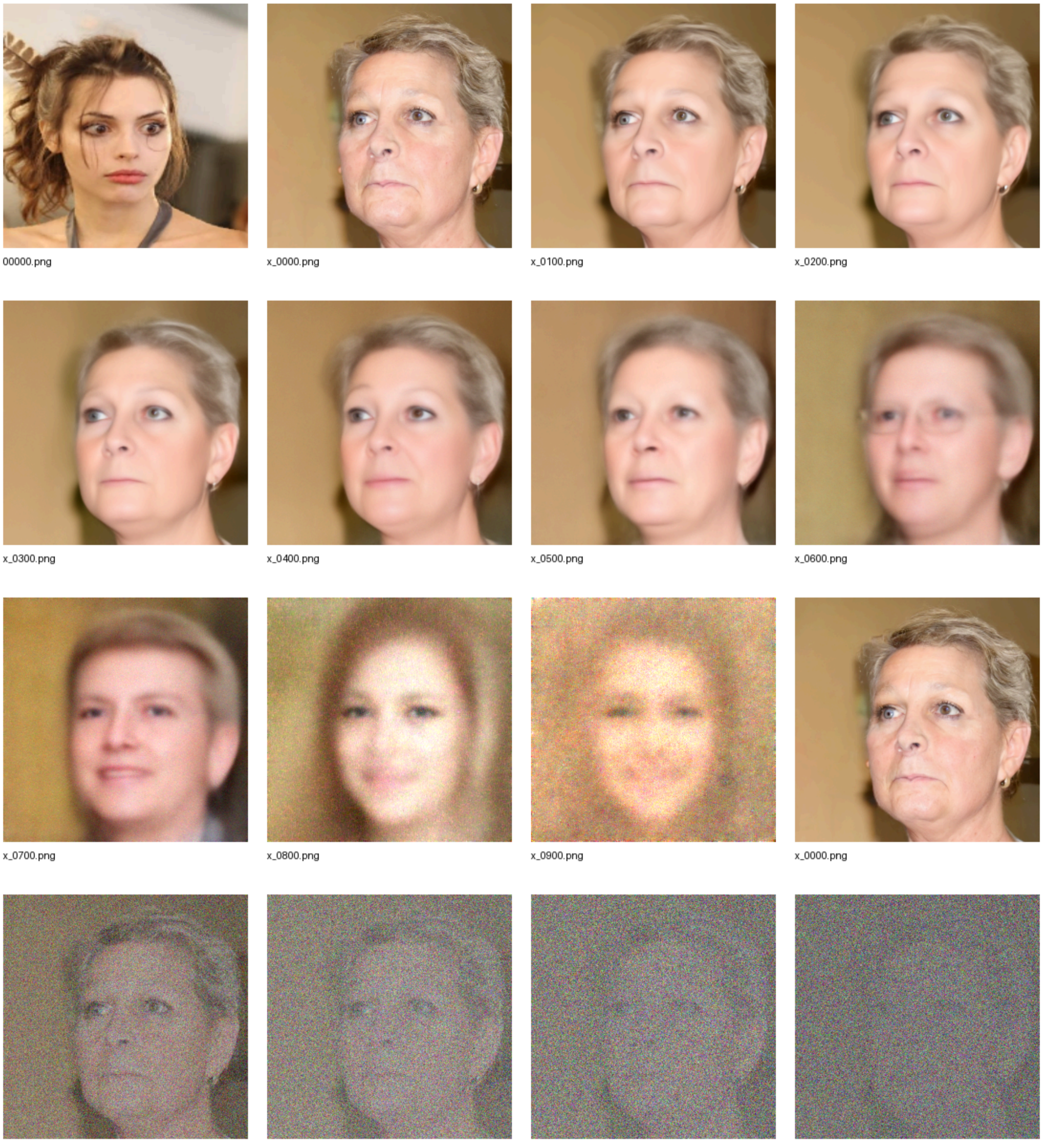

Figure: Trajectory of denoised samples generated by the GSS machinery with gradient guidance set to 0.

The saved trajectory shows the diffusion process moving from noisy states toward coherent face structure. The intermediate x_0 estimates become increasingly interpretable, with clear face-like structure appearing through the trajectory. This demonstrates that the GSS-only path is capable of producing meaningful images even though the leaked gradient is no longer used to guide the diffusion process.

The run produced 22 PNG files, including the target/label image, the final reconstruction, and sparse x_t and x_0 states at steps 900, 800, ..., 0.

As the diffusion process denoises the target image's noise, it produces a coherent image that is somewhat semantically similar to the target (female, face turned to the side, blonde, downturned mouth, shocked expression).
This is substantially better than the reconstruction experiments which used $m_r = 0.2$ and reconstructed only patches of color with no discernible facial features.

It is understandable that the reconstructed image does not exactly match the target, as we deliberately did not provide gradient-guidance to the diffusion model.

This gives the experiment its main conclusion.

The GSS sampling path can run successfully with

$m_r = 0$

and can produce natural, face-like outputs. Combined with the unconditional diffusion result, this makes two explanations less likely:

1. the FFHQ diffusion model is incapable of producing meaningful face images;
2. the basic GSS/diffusion sampling path fails simply because of the way it is implemented.

The remaining issue is therefore more specific.

When gradient guidance is enabled in the full GGSS-R reconstruction, the leaked victim gradient is supposed to steer the diffusion process toward the particular target image. The zero-guidance result shows that the sampler itself can reach the natural image manifold, but it does **not** show that the leaked gradient provides enough information to select the correct point on that manifold.

That is the key transition to the next diagnostic, direct_gradient_inversion_wo_diffusion.ipynb: remove diffusion altogether and ask whether the leaked gradient itself can reconstruct the target image. If direct gradient matching also produces candidates that are close in gradient space but far from the target in image space, the problem lies in the information carried by the leaked gradient.

Direct Gradient Inversion Without Diffusion

1. Motivation & Objective

This experiment follows the diffusion-model and GSS diagnostics and shifts the investigation away from the diffusion pipeline entirely.

The earlier unconditional_diffusion_manifold_diagnosis showed that the FFHQ diffusion model referenced by the authors can successfully denoise Gaussian noise into natural face images. The subsequent gss_with_no_guidance_diagnosis showed that the active GSS sampling path also remains on the natural image manifold when the leaked-gradient guidance is removed. Together, these results make a broken diffusion prior or diffusion-to-guidance execution path less likely to explain the failed GGSS-R reconstruction.

This leaves a more fundamental question:

Does the leaked victim gradient itself contain enough information to recover the target image?

GGSS-R assumes that guiding a candidate toward the target gradient also constrains it toward the image that produced that gradient. The previous experiments showed that the gradient objective can be optimized, but the reconstructed image still remains far from the target. To test whether this problem exists independently of diffusion, this experiment removes the diffusion model and GGSS-R sampling machinery completely and performs direct gradient inversion in image space.

The experiment therefore tests whether a candidate initialized from random noise can recover the target simply by minimizing its victim-gradient difference from the leaked target gradient. If direct inversion succeeds, the gradient contains sufficient information for reconstruction and the remaining problem would lie elsewhere in GGSS-R. If the gradient can instead be matched closely while the image remains substantially different from the target, then gradient matching itself is not sufficient to identify the target under this victim setup.

2. Methodology

The experiment reuses the exact single-epoch MLP_1 victim checkpoint and class-0 CelebA training-member target established in ggssr_unperturbed_reproduction_diagnostic. Those inputs are treated as read-only and copied into a separate experiment directory. No diffusion repository, diffusion checkpoint, or GGSS-R implementation is loaded.

The victim is the authors' three-layer fully connected model:

$3 \times 256 \times 256 \rightarrow 512 \rightarrow 128 \rightarrow 2$, with 100,729,730 parameters. The target is resized to 256 × 256, converted to a tensor, and normalized to [-1, 1], matching the preprocessing used in the previous reconstruction experiments. Its victim prediction is also verified as class 0 before the leaked target gradient is constructed.

Because `fc1.weight` alone contains more than 100 million gradient coordinates, matching the entire differentiable gradient at every optimization step would be unnecessarily expensive. The experiment therefore uses a fixed subset consisting of:

- 100,000 randomly selected `fc1.weight` coordinates;
- the complete `fc1.bias` gradient;
- the complete `fc2` gradients;
- the complete `fc3` gradients.

This gives **166,434 matched gradient coordinates** in total. The selected gradient is implemented analytically and verified against PyTorch autograd before reconstruction, ensuring that the optimized quantity is equivalent to the corresponding victim-model gradient.

For a candidate image x, the inversion minimizes the normalized squared gradient difference:

$$L(x) = \frac{\operatorname{mean}\left((g(x) - g_{\text{target}})^2\right)}{\operatorname{mean}\left(g_{\text{target}}^2\right) + 10^{-12}}$$

where,
x: Candidate image being optimized during inversion.
g(x): Gradient of the victim model's loss computed with respect to the candidate image x.
g_target: Leaked ground-truth gradient from the target image.
mean(·): Element-wise mean computed across all gradient components/parameters.
$(g(x) - g\_target)^2$: Element-wise squared difference between candidate and target gradients.
$mean(g\_target^2)$: Normalization term equal to the mean squared magnitude of the target gradient.
10^{-12}: Small numerical stability constant (epsilon) to prevent division by zero.

Rather than optimizing unrestricted pixel values directly, a free tensor z is optimized and converted to the candidate image using

x = tanh(z),

which keeps every reconstructed pixel within the victim's expected [-1, 1] input range.

A single 1,500-step Adam run is first performed as a sanity check. The main experiment then repeats the inversion across **eight independent random restarts**, each using:

- 1,500 optimization steps;
- learning rate 0.05;
- the same fixed target gradient and selected coordinates;
- a different reproducible random initialization.

Gradient-space progress is measured using Euclidean gradient distance, relative gradient distance, and cosine similarity. Pixel MSE against the actual target is measured separately. This distinction is central to the experiment: the question is not only whether the leaked gradient can be matched, but whether matching it actually reconstructs the image that produced it.

3. Results & Observations

The direct gradient-inversion objective works as intended. In the initial sanity run, relative gradient distance decreases from **1.1363** to **0.5106**, while gradient cosine similarity increases from **0.4039** to **0.8598**. The optimizer is therefore clearly capable of moving the candidate toward the leaked target gradient without any diffusion machinery.

The corresponding improvement in image space is much smaller. Pixel MSE begins at **0.6683** and remains **0.6206** after 1,500 steps. The candidate becomes substantially more similar to the target in gradient space without becoming a successful reconstruction of the target image.

The eight independent restarts make this result considerably stronger.

Figure: Final reconstruction of eight independent restarts under direct gradient inversion attack.

The best gradient matches are runs 4 and 7. Their final relative gradient distances are only **0.0412** and **0.0433**, with cosine similarities of **0.99915** and **0.99906**. The reconstructed gradients are therefore almost perfectly aligned in direction with the leaked target gradient.

Despite this, their pixel MSEs remain **0.4311** and **0.4516**. None of the eight random restarts produces a low-error recovery of the target image. All eight final candidates are classified as class 0 by the victim, showing that the optimization can satisfy the target's class-related behavior without recovering its image identity.

Run	Seed	Final Loss	Final Gradient Distance	Final Relative Gradient Distance	Final Gradient Cosine	Final Pixel MSE	Final Victim Prediction
0	3026	0.0796	0.3748	0.2822	0.9593	0.4580	0
1	3027	0.0806	0.3771	0.2840	0.9588	0.4627	0
2	3028	0.2808	0.7037	0.5299	0.8480	0.6331	0
3	3029	0.2552	0.6710	0.5052	0.8630	0.6217	0
4	3030	0.0016	0.0547	0.0412	0.9991	0.4311	0
5	3031	0.0784	0.3720	0.2801	0.9599	0.4569	0
6	3032	0.2287	0.6352	0.4782	0.8782	0.6115	0

7	3033	0.0018	0.0575	0.0433	0.9990	0.4515	0

Table: Gradient-space and image-space metrics of final reconstructed images across eight independent runs of direct gradient inversion.

The best gradient matches are also the best two reconstructions by pixel MSE, so gradient matching is not completely unrelated to image similarity. Moving closer to the target gradient can move the candidate in a useful image-space direction. The important result is that this relationship is not strong enough for reconstruction: even an almost exact gradient-direction match still corresponds to a substantially different image.

This rules out an important explanation for the earlier failure. The reconstruction problem is not caused solely by the presence of the diffusion model or by GGSS-R's diffusion sampling procedure. The same separation between gradient similarity and image recovery remains when diffusion is removed and the leaked gradient is attacked directly.

However, this experiment alone does not establish why that separation occurs or prove that the target gradient is numerically non-unique. It establishes the empirical problem: direct optimization can produce very strong gradient matches without recovering the target.

That result motivates gradient_identifiability_analyses, which examines the victim gradients across a larger image population, compares gradient-space similarity with image-space similarity, and analyzes the victim layer by layer to determine what information the leaked gradient is actually carrying.

Gradient Identifiability Analyses

1. Motivation & Objective

This experiment is the final diagnostic in the reconstruction sequence. It investigates **why minimizing the leaked-gradient difference has repeatedly failed to recover the target image**.

The preceding experiments progressively isolated the main components of GGSS-R.
unconditional_diffusion_manifold_diagnosis established that the diffusion model can generate natural face images from noise.
gss_with_no_guidance_diagnosis showed that the active GSS sampling path also remains capable of producing images on the natural manifold without gradient guidance, even though the reconstructed image is **not** similar to the target due to the absence of gradient guidance.
direct_gradient_inversion_wo_diffusion then removed diffusion entirely and optimized an image directly against the leaked victim gradient.

That direct inversion produced the key motivation for this experiment. Some reconstructed candidates achieved relative gradient distances near **0.04** and gradient cosine similarities above **0.999**, yet still remained far from the target in pixel space. The problem therefore appears deeper than diffusion: **a candidate can become extremely close to the leaked gradient without recovering the image that produced it.**

This experiment examines whether the single-epoch MLP_1 victim gradient is sufficiently image-specific for reconstruction in the first place.

A useful leaked gradient must do more than vary across inputs or encode the victim's binary Smiling classification. For image reconstruction, gradient similarity must meaningfully constrain image similarity, and the gradient must retain enough target-specific information to distinguish the original image from alternative inputs.

The experiment therefore studies gradient identifiability at three levels:

- how much individual gradient coordinates vary across different images;
- how gradient-space relationships correspond to class and image-space relationships across a population;
- where target-specific gradient information is distributed across the victim model's layers.

The goal is not to assume in advance that the gradient is non-identifying. It is to determine what the measured gradient structure can actually tell us about the reconstruction failures observed throughout the preceding experiments.

2. Methodology

The experiment uses the same single-epoch MLP_1 victim checkpoint from ggssr_unperturbed_reproduction_diagnostic. The original GGSS-R target and saved reconstruction trajectory are also reused where target-specific comparisons are required.

The victim contains **100,729,730 parameters**, making it impractical to retain complete gradients for hundreds of images simultaneously. The analysis therefore combines memory-efficient population representations with exact full-gradient calculations where they are most important.

A balanced population of **200 CelebA images** is sampled using a fixed seed:

- 100 images with Smiling == 0;
- 100 images with Smiling == 1.

Every image is passed through the same victim and differentiated using the same classification objective.

For per-coordinate analysis, **100,000 fixed gradient coordinates** are sampled and observed across all 200 images. Their variance and sign entropy are measured to determine how strongly individual gradient elements change across inputs.

Per-Coordinate Sample Variance:

$$\sigma_k^2 = \frac{1}{N-1}\sum_{n=1}^{N}\left(g_{n,k} - \bar{g}_k\right)^2$$

where,

- **N:** Total number of sampled images (N = 200).
- **g_{n, k}:** Gradient value of the k-th parameter coordinate for image n.
- **ḡ_k:** Mean value of the k-th gradient coordinate across all N images.

Coordinate Sign Entropy:

$$H_k = -p_k \log_2(p_k) - (1-p_k)\log_2(1-p_k)$$

where,

p_k: Fraction of images in the population for which gradient coordinate k is strictly positive (g_(n,k) > 0).

H_k: Binary sign entropy measuring directional stability of coordinate k (0 = identical sign across all images; 1 = maximum sign instability).

For population-wide pairwise analysis, each full gradient is compressed into the same **256-dimensional random projection**. Each projected dimension is constructed from 4,096 gradient coordinates, so the projection samples information from approximately one million coordinates overall. This compact representation makes it possible to perform pairwise comparisons, PCA, clustering, and nearest-neighbour analysis without storing the complete 100-million-dimensional gradient population.

Two gradient-space measures are used:

- Euclidean distance, which measures absolute separation;
- cosine similarity, which measures directional agreement;

The 200 projected gradients are then analyzed in several complementary ways.

First, same-class and different-class image pairs are compared to determine whether the gradient contains class-related structure.

PCA and K-means clustering are then used to characterize broader population structure and redundancy.

Principal Component Analysis:

$$v_1 = \arg\max_{\|v\|_2=1} \sum_{n=1}^{N} \left((\hat{g}_n - \bar{\hat{g}}) \cdot v\right)^2$$

where,

ĝ_n: Projected gradient vector of image n.

ḡ̂: Population mean vector across all 200 projected gradients.

v_1: First principal component direction capturing maximum gradient variance.

K-Means Clustering:

$$J_{\text{K-Means}} = \sum_{k=1}^{K} \sum_{n \in C_k} \|\hat{g}_n - \mu_k\|_2^2$$

where,

K: Number of target clusters (K = 2).

C_k: Set of projected gradient indices belonging to cluster k.

μ_k: Centroid (mean vector) of cluster k in projected gradient space.

Gradient-space nearest neighbours test whether images with highly similar gradients necessarily share the same class or image characteristics.

The experiment next computes pixel MSE for all **19,900 (200C2) unique image pairs** and directly compares image-space distance with gradient-space similarity. Spearman correlations quantify whether images that are closer in gradient space also tend to be closer in image space.

Image-space Pixel MSE:

$$\mathrm{MSE}_{\mathrm{pixel}}(x_a, x_b) = \frac{1}{H \cdot W \cdot C} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{c=1}^{C} (x_{a,i,j,c} - x_{b,i,j,c})^2$$

where,

x_a, x_b: Raw pixel tensors for images a and b.

H, W, C: Image spatial dimensions and channels (Height = 256, Width = 256, Channels = 3).

Spearman Rank Correlation:

$$\rho_s = 1 - \frac{6 \sum_{k=1}^{P} d_k^2}{P(P^2 - 1)}$$

where,

P: Total number of unique image pairs (P = 19,900 from 200 choose 2).

d_k: Difference between the rank of pair k in gradient similarity versus its rank in pixel MSE.

The original GGSS-R target is then reintroduced for a more exact analysis.
Unlike the population-wide projected analysis, its gradient is compared with all 200 sampled alternatives using the **full victim gradient**. This checks whether images nearest to the target in exact gradient space are also visually close to it.

Finally, the target comparison is decomposed across the victim's six parameter tensors: fc1.weight, fc1.bias, fc2.weight, fc2.bias, fc3.weight, fc3.bias.

Layer-Wise Exact Gradient Distance:

$$D_{\mathrm{rel}}^{(l)}(g, g_{\mathrm{target}}) = \frac{\|g^{(l)} - g_{\mathrm{target}}^{(l)}\|_2}{\|g_{\mathrm{target}}^{(l)}\|_2}$$

l: Specific parameter tensor index of the victim model (e.g., fc1.weight, fc2.weight).

g^(l): Candidate image gradient corresponding to parameter tensor l.

g_target^(l): Target image gradient corresponding to parameter tensor l.

This is especially important for MLP_1. Its fc1.weight tensor contains **100,663,808 parameters**, or about **99.9% of the entire model**, and directly receives the flattened image as input.
The layer-wise analysis therefore tests whether this extremely large input-facing layer actually carries a strong target-dependent gradient signal or whether the leaked gradient is dominated by the much smaller downstream classification layers.

The saved GGSS-R reconstruction trajectory is also evaluated against the target in both image and exact gradient space.

This connects the population-level diagnosis back to the reconstruction failure that motivated the experiment.

3. Results & Observations

The results show that the victim gradient is **not devoid of image information**, but the information it contains is **not sufficiently tied to image identity** to make gradient matching equivalent to image recovery.

At the individual-coordinate level, gradient variation is limited across much of the sampled population. Among the 100,000 inspected coordinates, the median variance is only **6.28×10^{-8}** and the median sign entropy is **0.242 bits**. Only **10.1%** of coordinates have sign entropy above 0.9 bits. Most sampled coordinates therefore change relatively little or retain a stable sign across different images, while stronger variation is concentrated in a much smaller subset.

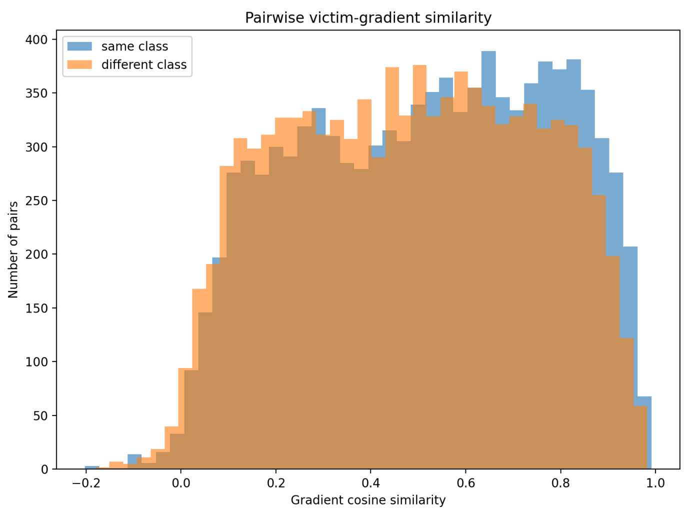

Figure: Gradient cosine distribution across the two classes.

The projected population nevertheless contains real structure. Same-class image pairs have a slightly higher mean gradient cosine similarity than different-class pairs (**0.5165 vs. 0.4819**) and a lower mean projected-gradient distance (**0.00143 vs. 0.00196**). The gradients therefore contain some information related to the victim's Smiling classification, but the same-class and different-class distributions overlap substantially.

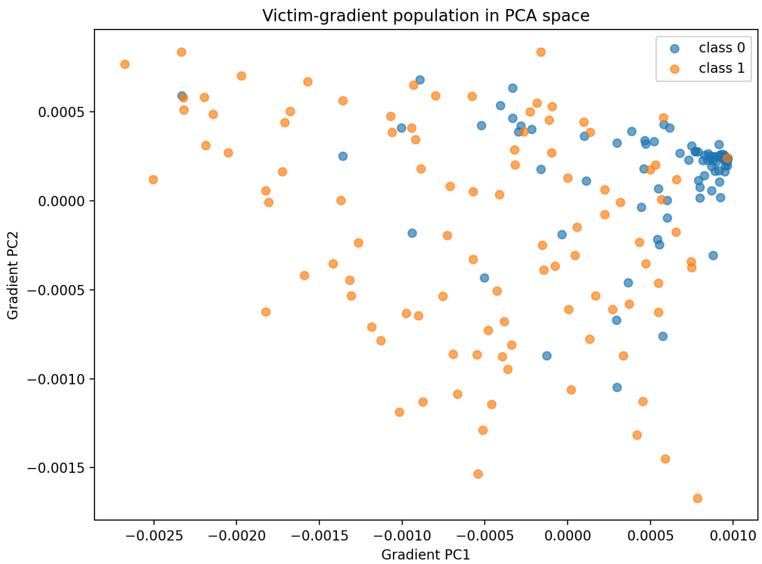

Figure: Victim gradient population in PCA subspace.

The PCA results show that this projected gradient population is also highly redundant: **16 components explain 95%** of the variation in the 256-dimensional representation, and 39 explain 99%.

K-means finds its best tested silhouette score at two clusters (**0.4035**), indicating broad population structure but not cleanly separated groups. Because the clustering is unsupervised and is not matched directly to the labels, this does not establish that the two clusters correspond to the two Smiling classes.

The nearest-neighbour analysis further shows that high gradient similarity is not simply class identity. For one class-0 query, the closest gradient-space neighbour is a **class-1** image with cosine similarity 0.9713. Gradient space therefore captures input-dependent structure beyond the binary label, but a highly similar gradient does not uniquely determine even the image's class, much less its exact visual identity.

The population-wide image comparison gives a more nuanced result. Gradient cosine similarity has a substantial negative Spearman correlation with pixel MSE:

$\rho = -0.6348$.

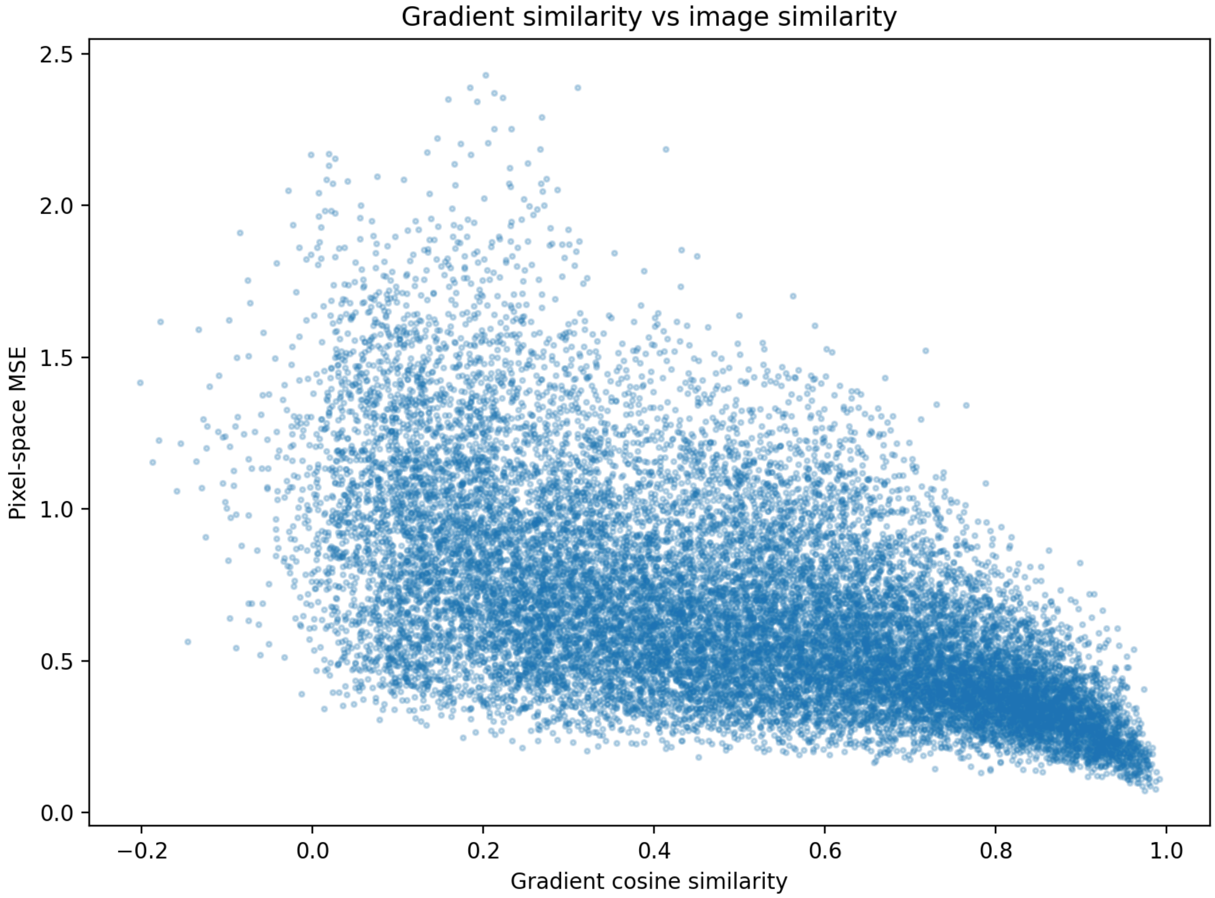

Figure: Gradient cosine vs Pixel MSE.

Images with more directionally similar gradients therefore tend, in general, to be more similar in pixel space. Gradient information is not arbitrary or disconnected from image content.

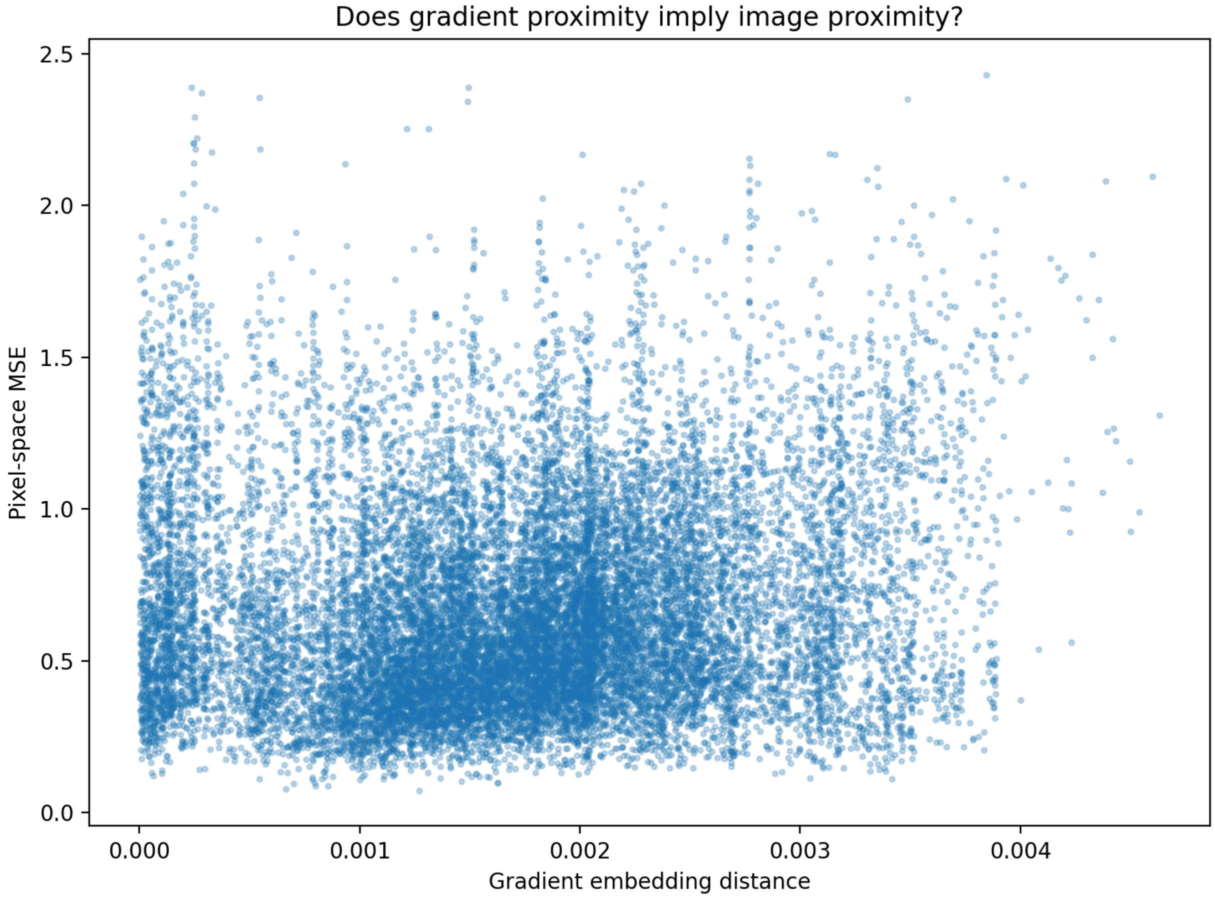

Figure: Gradient distance vs Pixel MSE.

Raw gradient Euclidean distance is much less informative, with

$\rho = 0.1250$

against pixel MSE.

Directional gradient agreement is therefore much more closely associated with image similarity than absolute projected-gradient distance in this population. This can be observed from the figures as well: High cosine similarity values have a tight / narrow clustering of points, meaning that their corresponding pixel MSEs do not deviate much; while low distance values show haphazardly scattered points, so their corresponding pixel MSEs vary between a large range. This suggests that maximizing the gradient cosine might have a higher chance of recovering images with low image-space error than minimizing gradient distance does.

This population-level relationship is however not a one-to-one mapping. The exact target comparison shows that images closest to the target in full-gradient Euclidean distance or even cosine similarity can still have large pixel errors.

The closest of the 200 sampled alternatives has full-gradient distance **1.1422**, but its pixel MSE from the target is still **0.6008**. The sampled population does not contain another image with an effectively identical full gradient, so it is not that gradients are completely non-unique. What they establish is that gradient closeness is not a reliable proxy for visual identity within the tested population under the current victim model setup.

The layer-wise analysis provides the strongest explanation for why this problem occurs under the current victim model.

Although `fc1.weight` contains approximately **99.9% of all victim parameters** and is the layer most directly connected to the input pixels, its gradient shows very weak target-relative directional structure across the population. Its mean cosine similarity with the target gradient is only **0.0503**. In contrast, the much smaller `fc3.weight` has mean target cosine **0.6024**, while `fc2.weight` and `fc3.weight` dominate the raw target-gradient differences.

This means that the leaked gradient signal is distributed very unevenly. The enormous input-facing layer that has the greatest capacity to retain image-specific information contributes relatively little useful gradient signal, while the smaller downstream layers associated more directly with the classification decision contribute much more strongly to the gradient objective.

The victim's behavior is consistent with this finding. The single-epoch model already classifies the target confidently and produces a very small target loss. The measured layer-wise gradients indicate that the backpropagated signal has become extremely weak by the time it reaches `fc1`. This provides a plausible mechanism for the observed identifiability problem: the model can be a successful `Smiling` classifier while its leaked gradient carries relatively little signal through the layer that is most directly tied to individual pixels.

The saved GGSS-R trajectory confirms the practical consequence. As reconstruction progresses, gradient and image metrics do improve together: gradient distance falls from **0.8918** to **0.2898**, cosine similarity rises to **0.9764**, and pixel MSE falls from **0.8997** to **0.6085**. Gradient guidance is therefore moving the reconstruction in a direction that is somewhat informative about the target.

But the improvement saturates far from actual recovery. Even with cosine similarity of **0.9764**, the final saved reconstruction still has MSE **0.6085** and LPIPS **0.7490**. This is the same pattern observed more strongly in direct gradient inversion, where cosine similarity could exceed 0.999 without producing a recognizable target reconstruction.

Taken together, the diagnostic sequence points to a specific limitation of this reproduction setup. The diffusion model can generate natural images, the GSS execution path can operate on that manifold, the gradient objective can propagate through the candidate image, and direct gradient optimization can achieve extremely strong gradient alignment. The remaining failure is that **strong gradient alignment under the single-epoch `MLP_1` victim does not constrain the candidate tightly enough to recover the original image**.

The results therefore shift the next experimental question from the GGSS-R sampler to the victim model itself. A useful follow-up is to test whether a victim architecture or training state that preserves a stronger gradient signal in its input-facing layers produces a tighter relationship between gradient similarity and image identity, and consequently enables better reconstruction.

Victim Model Comparison for Reconstruction

1. Motivation & Objective

The previous gradient_identifiability_analyses experiment narrowed the reconstruction failure to the victim model itself.

Earlier diagnostics had already shown that the diffusion model can generate natural face images, that the GSS execution path remains on the natural image manifold, and that the leaked-gradient objective can be optimized successfully. Direct gradient inversion then showed that a candidate could become extremely close to the leaked gradient without recovering the target image. The identifiability analysis traced this behavior to weak image-specific gradients reaching the input-facing layer of the authors' single-epoch MLP_1.

This experiment asks whether reconstruction quality changes when the **victim architecture** and **training state** are changed. The objective is to perform a quantitative analysis of reconstruction susceptibility to not only model architecture, but also model training state.

Two model states are compared:

- **epoch 0**: the randomly initialized model before any training;
- **epoch 1**: the same model after one epoch of CelebA Smiling classification training.

Epoch 0 is useful because it separates gradient identifiability from learned classifier confidence. An untrained model has not yet learned the classification task, so its loss can remain substantial. However, a large loss alone does not guarantee useful image gradients: that error signal must still survive backpropagation through the network and reach layers that retain information about the input.

Four custom architectures are tested:

- mlp_small: a **25.2M-parameter MLP**. MLP victims are reported as relatively reconstructable in GGSS-R, and the first fully connected layer has a direct pixel-to-weight relationship with the input.
- lenet_sigmoid: a severely under-parameterized **34K-parameter CNN**, used to test whether limited capacity affects the amount of image-specific information available in the gradient.
- cnn_gi: a moderate **4.3M-parameter CNN**, representing a middle ground in capacity.
- cnn_overparam: a large **35.1M-parameter CNN**, used to examine whether substantially increasing convolutional model capacity improves (by memorizing image-specific identity) or worsens (by diminishing the loss) gradient identifiability.

These architectural choices are hypotheses to test rather than expected outcomes. The reconstruction results determine which model states actually expose enough image-specific gradient information for inversion.

The experiment uses direct gradient inversion rather than GGSS-R. The target gradients are completely unperturbed, allowing the effect of victim architecture and training state to be isolated without introducing diffusion sampling or gradient perturbation.

This does not make diffusion guidance unnecessary. GGSS-R is designed for the harder setting where leaked gradients may be perturbed. Direct inversion can reproduce noisy features of that perturbation because it only tries to satisfy the observed gradient. A diffusion prior instead constrains the candidate toward the natural image manifold, which can make reconstruction more robust to noisy gradients. The purpose here is simply to first establish which clean victim gradients are identifiable.

2. Methodology

The same class-0 CelebA training-member target used in the preceding reconstruction diagnostics is retained. Images are resized to 256×256, converted to tensors, and normalized to [-1, 1].

Victim architectures

`mlp_small` contains one 128-unit hidden layer:

$3 \times 256 \times 256 \rightarrow 128 \rightarrow 2.$

Its first fully connected layer accounts for almost all of its 25,166,210 parameters.

`lenet_sigmoid` contains four sigmoid convolutional layers followed directly by a two-class fully connected layer. It contains only 34,022 parameters.

`cnn_gi` contains five sigmoid convolutional layers followed by a 128-unit fully connected hidden layer and a two-class output layer, totaling to 4,334,114 parameters.

`cnn_overparam` follows a similar structure with substantially wider convolutional layers and a 512-unit hidden layer, giving 35,106,946 parameters.

For each architecture, the randomly initialized weights are first saved as the epoch-0 checkpoint. The same model is then trained for one epoch using cross-entropy loss and Adam with learning rate 0.001.

After one epoch, the training outcomes are:

Model	Parameters	Training loss	Training accuracy
mlp_small	25,166,210	0.5445	73.38%
lenet_sigmoid	34,022	0.7099	51.34%
cnn_gi	4,334,114	0.7046	51.53%
cnn_overparam	35,106,946	0.7186	50.67%

Only mlp_small learns the classification task substantially within the single epoch. The three CNNs remain close to chance-level accuracy.

Target-gradient construction

For every architecture and epoch, the target image is passed through the victim and its classification loss is differentiated with respect to every model parameter:

$$g_{\text{target}} = \nabla_\theta \mathcal{L}(f_\theta(x_{\text{target}}), y_{\text{target}})$$

The gradient norm of every parameter tensor is also recorded. This is important because the total gradient magnitude alone does not show whether a useful signal reaches the layers that interact most directly with the image.

Direct gradient inversion

Each reconstruction begins from a random image. The candidate is passed through the same frozen victim and differentiated using the known target label:

g(x) = ∇_θ L(f_θ(x), y_target)

The candidate image is then optimized to minimize the mean squared difference between the complete candidate and target gradients:

L_inv = MSE(g(x), g_target)

The gradient computation remains differentiable so that this objective can be backpropagated through the victim and into the candidate image.

Each architecture × epoch combination is tested using three independent random seeds, giving:

4 architectures × 2 epochs × 3 seeds = 24 reconstruction runs.

Every run uses:

- 1,500 Adam optimization steps;
- learning rate 0.1;
- the complete, unperturbed victim gradient;
- candidate clipping to [-1, 1].

Gradient-space progress is tracked using relative gradient distance and cosine similarity. Image reconstruction is evaluated independently using pixel MSE, PSNR, and LPIPS. The full optimization trajectories are retained so that changes in gradient space can be compared directly with changes in image space.

3. Results & Observations

Reconstruction results

The full architecture × epoch × seed sweep produces only one clear successful condition: `mlp_small` at epoch 0.

Model state	Mean MSE	Mean PSNR	Mean LPIPS
`mlp_small`, epoch 0	**0.00113**	**29.72 dB**	**0.1189**
`mlp_small`, epoch 1	0.16397	7.88 dB	1.2020
`lenet_sigmoid`, epoch 0	0.15109	8.21 dB	1.2704
`lenet_sigmoid`, epoch 1	0.12478	9.04 dB	1.2781

cnn_gi, epoch 0	0.15194	8.18 dB	1.2714
cnn_gi, epoch 1	0.15194	8.18 dB	1.2714
cnn_overparam, epoch 0	0.15194	8.18 dB	1.2714
cnn_overparam, epoch 1	0.15194	8.18 dB	1.2714

Table: Image-space metrics of final reconstructions found by four model architectures, each at two training states.

Across all three seeds for mlp_small at epoch-0, the initially random image gradually converges toward the target in both gradient space and image space. The three successful runs finish with MSEs of 0.00154, 0.00116, and 0.00068. Their PSNR values reach 28.12, 29.37, and 31.67 dB, while LPIPS falls to 0.164, 0.127, and 0.066.

Step	**Gradient Distance**	**Relative Gradient Distance**	**Gradient Cosine**	**MSE**	**PSNR**	**LPIPS**
0	22.1732	1.0375	0.0375	0.1513	8.2023	1.2787
199	21.3166	0.9974	0.0984	0.1475	8.3133	1.2761
399	21.2970	0.9965	0.1022	0.1439	8.4199	1.2744
599	21.2411	0.9939	0.1173	0.1378	8.6070	1.2718

799	21.0238	0.9837	0.1810	0.1247	9.0405	1.2656
999	19.8097	0.9269	0.3754	0.0930	10.3153	1.2419
1199	13.6020	0.6364	0.7713	0.0340	14.6801	0.9669
1399	5.1932	0.2430	0.9700	0.0046	23.3692	0.3731
1499	2.9190	0.1366	0.9906	0.0015	28.1164	0.1640

Table: Reconstruction trajectory of mlp_small at epoch 0. (Seed 0).
As the gradient distance is optimized, image reconstruction improves consistently in tandem. This shows that the gradient similarity to image similarity mapping does exist for particular architectures at specific epochs.

The trajectories show the same pattern. Relative gradient distance falls from approximately 1.0 to 0.083–0.137, cosine similarity rises from approximately 0.04 to 0.991–0.997, and image MSE falls from approximately 0.151 to near zero. The successful reconstruction is therefore consistent across all three random initializations.

However, the same architecture of mlp_small behaves very differently at epoch 1. All three runs fail to reconstruct the target. Two remain essentially at their starting image-space error, while one becomes worse. Mean MSE rises to 0.164, PSNR falls below 8 dB, and LPIPS remains above 1.2.

Figure: Final image reconstructed by the same architecture mlp_small at different training states. Left: mlp_small at epoch 0. Right: mlp_small at epoch 1. Seed 0.

On the other end of the spectrum, for the under-parameterized CNN lenet_sigmoid, the model state at epoch 0 is not able to reconstruct the image. Its MSE only changes from approximately 0.152 to 0.151. However, its epoch-1 run reduces MSE to approximately 0.125, and is able to reconstruct the image semantics to some extent, even though the pixels themselves stay noisy.

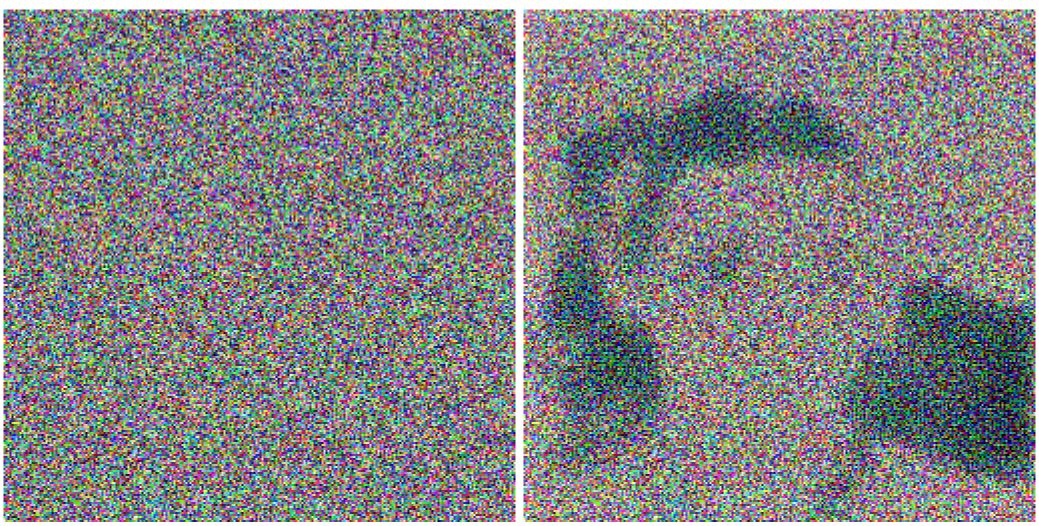

Figure: Final image reconstructed by the same architecture lenet_sigmoid at different training states. Left: lenet_sigmoid at epoch 0. Right: lenet_sigmoid at epoch 1. Seed 0.

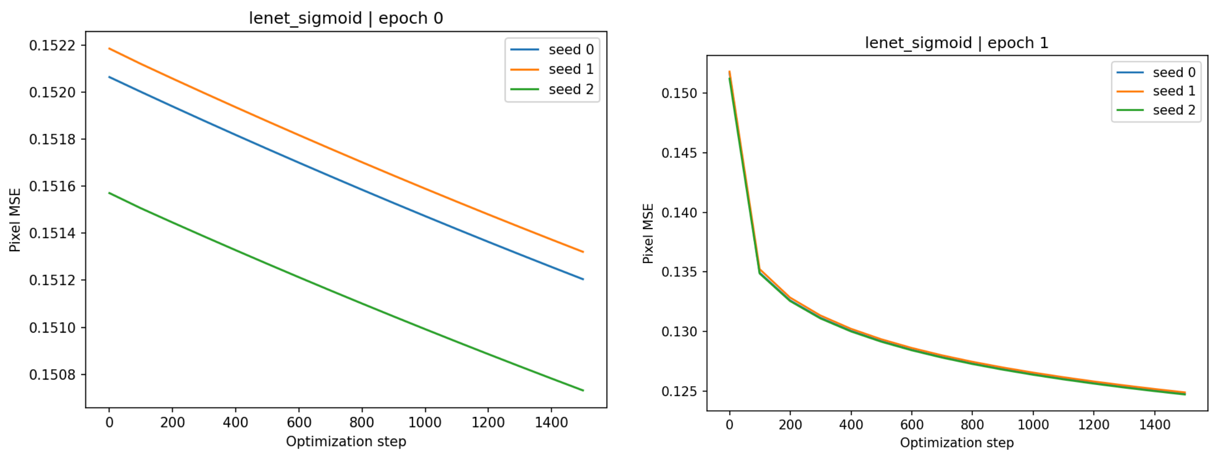

Figure: MSE trajectory of images reconstructed by lenet_sigmoid at epoch 0 vs at epoch 1. lenet_sigmoid performs better reconstruction at its more-trained model state.

This once again demonstrates how reconstruction vulnerability is dependent on training state, and that retaining the same model architecture alone will not ensure that reconstruction can be performed with the same efficacy across varying gradient states. Success in reconstruction is determined by whether the loss can be differentiated all the way back into the input-facing layer. If the gradient vanishes during backpropagation during the later layers, and no meaningful gradient is able to reach the earlier/input layers that are directly connected to the image, then the pixels have no way of being optimized.

As for the two CNN architectures cnn_gi and cnn_overparam, they fail to produce useful reconstructions altogether. Both epoch 0 and epoch 1 trajectories are completely flat: pixel MSE remains around 0.152 for all 1,500 steps.

Main interpretations

The layer norms explain why the untrained mlp_small is vulnerable to reconstruction.

Model	**Epoch**	**Parameter**	**Numel**	**Gradient Norm**
mlp_small	0	fc1.weight	25,165,824	21.0519
mlp_small	0	fc1.bias	128	0.0905

mlp_small	0	fc2.weight	256	3.6239
mlp_small	0	fc2.bias	2	0.6526
mlp_small	**0**	**TOTAL**	**25,166,210**	**21.3717**
mlp_small	1	fc1.weight	25,165,824	3.8948
mlp_small	1	fc1.bias	128	0.0167
mlp_small	1	fc2.weight	256	2.4679
mlp_small	1	fc2.bias	2	0.2987
mlp_small	**1**	**TOTAL**	**25,166,210**	**4.6206**

At epoch 0, its target loss is 0.6189 and its total gradient norm is 21.37.
Almost all of that useful signal reaches the first layer: $||\nabla W\_fc1|| = 21.05$

For a fully connected first layer $z\_1 = W\_1 x + b\_1$
the weight gradient for a single input has the form $\partial L / \partial W\_1 = \delta\_1 x^T$.

$$\frac{\partial \mathcal{L}}{\partial W_1} = \delta_1 x^T$$

The image therefore appears directly within the first-layer gradient whenever the backpropagated error δ_1 remains sufficiently strong.

That condition is present in the untrained MLP. Its gradient objective provides a strong image-specific direction, and the candidate moves steadily toward the target.

After one epoch of training, the mlp_small target loss falls from 0.6189 to 0.2372.

Its total target-gradient norm simultaneously falls: 21.37 → 4.62
while the first-layer weight-gradient norm drops even more clearly: 21.05 → 3.89

The reconstruction quality collapses at the same time.

For this model, the result matches the behavior identified in the previous gradient_identifiability_analyses: as the victim becomes more confident on the target, the classification loss and backpropagated error become smaller, leaving much less input-specific gradient signal for inversion.

The epoch-0 versus epoch-1 MLP comparison therefore shows that **training state alone can substantially change reconstruction vulnerability even when the architecture is unchanged**. The authors' paper did **not** explicitly analyse varying training states across the same model architecture.

They established **Reconstruction Vulnerability** as a metric of how susceptible models are to gradient inversion / data reconstruction attacks, and they showed a comparison of model architectures with their RV values, with the 3-layered MLP being the easiest to reconstruct from. However, they left unexamined the question of how training progression within a single architecture alters reconstruction vulnerability. This experiment thus directly answers that question.

Additional observations

The CNN results show that an untrained model is not automatically reconstructable.
Their epoch-0 target losses are all substantial:

- lenet_sigmoid: 0.6504;
- cnn_gi: 0.6476;
- cnn_overparam: 0.9045.

Despite this, the gradient reaching the early feature layers is extremely weak.

For lenet_sigmoid, the total epoch-0 gradient norm is actually the largest among the four models: 38.02.

However, almost all of this norm is concentrated in the final classifier: $||\nabla W_{fc}|| = 38.01$
The first convolutional layer has a weight-gradient norm of only: 0.00150

The same pattern is stronger in the deeper CNNs.

For cnn_gi at epoch 0: $||\nabla g||_{total} = 9.65$

but $||\nabla W\_conv1|| \approx 1.0 \times 10^{-5}$

For cnn_overparam: $||\nabla g||\_total = 18.39$
while $||\nabla W\_conv1|| \approx 1.5 \times 10^{-5}$

Most of their gradient energy instead appears in the later fully connected layers.

This shows that for an image to be optimized, the error signal must survive the complete backpropagation route and remain informative in layers connected to the image.

This experiment also resolves the central question raised by gradient_identifiability_analyses.

Clean victim gradients **can** contain enough information for high-quality image reconstruction. The three successful mlp_small epoch-0 runs demonstrate this directly. The success of gradient identifiability and gradient-to-image mapping however depends strongly on **both the victim architecture and its state**. These two together determine whether a gradient can remain sufficiently strong and image-specific in the parts of the network that retain information about the original input.

GGSS-R Scheduled Guidance Rate for Perturbed Gradient Reconstruction

1. Motivation & Objective

GGSS-R reconstructs private images by coupling gradient guidance with reverse diffusion. The diffusion model acts as a natural-image prior, while the leaked gradient guides the generated image toward the private target.

The paper uses a fixed guidance rate m_r to balance these two components and reports **m_r = 0.2** as the best static setting.

$$d_m = d^{\text{sample}} + m_r(t) \cdot (d^* - d^{\text{sample}})$$

where, d_sample is the unconditional denoising direction,
d* is the optimal gradient-guided direction,
m_r is the control,
and d_m is the gradient-guided conditional diffusion direction.

The paper's noisy-gradient experiments, however, show an important limitation of keeping that balance unchanged throughout reconstruction. With moderate Gaussian perturbation, reconstruction quality can reach a strong intermediate state and then deteriorate during later diffusion steps. Because the leaked gradient itself contains noise, continuing to enforce it strongly can make the reconstruction increasingly follow the noise-contaminated gradient rather than the clean target.

Early stopping does not provide a practical solution: an attacker does not have the private target image and therefore cannot know when the best reconstruction has been reached.

This experiment extends GGSS-R by making the guidance rate **timestep-dependent**. The reconstruction begins with strong gradient guidance to steer the sample toward the target, then progressively reduces that influence so that the diffusion prior has more control during later refinement.

The paper's fixed setting, m_r = 0.2,

is compared with a linear schedule, m_r(i) = 0.2 × (i / 999),

for reverse-diffusion indices i = 999, ..., 0. The scheduled run therefore starts at m_r = 0.2 and ends at m_r = 0.

The experiment specifically tests reconstruction from a gradient perturbed with **Gaussian noise variance 0.01**. The question is whether decaying m_r can preserve useful early gradient guidance while reducing the late-stage influence of the perturbation.

2. Methodology

The experiment retains the authors' GGSS-R reconstruction pipeline and their referenced FFHQ diffusion checkpoint.

The victim is the paper's **MLP-3 architecture**, called MLP_1 in the authors' implementation, with approximately 100.7M parameters. It is used at **epoch 0**, directly after initialization, so that the experiment starts from a victim whose clean gradient is suitable for reconstruction rather than introducing weak gradients caused by training.

A CelebA target is prepared at the 256×256 resolution required by the victim and reconstruction pipeline. Its clean victim gradient g is calculated and perturbed before reconstruction:

$$\tilde{g} = g + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.01)$$

Both experimental conditions receive exactly the same perturbed gradient.

Two 1000-step GGSS-R reconstructions are then run. The **static condition** uses m_r = 0.2 at every reverse-diffusion step. The **scheduled condition** begins at the same value and linearly decreases m_r to zero over the trajectory.

The scheduling is introduced through a small hook in the authors' GSS implementation rather than by changing the underlying gradient-conditioning procedure. Both runs use the same target, victim initialization, diffusion checkpoint, noisy gradient, and initial diffusion noise x_T. This keeps the guidance-rate policy as the experimental difference.

Reconstruction quality is measured directly against the target using:

- **MSE**, where lower is better;
- **PSNR**, where higher is better;
- **LPIPS**, where lower is better.

Metrics are recorded throughout the full trajectory rather than only at the final output. Sparse reconstruction snapshots, metric curves, comparison figures, and a 10×2 contact sheet are also generated to compare how the two reconstructions evolve.

Image-space metrics are used as the measure of reconstruction quality because the attack is deliberately matching a perturbed gradient. A closer match to that noisy gradient is not necessarily a closer reconstruction of the clean target.

3. Results & Observations

The scheduled guidance run produces the better reconstruction, with the advantage becoming progressively clearer during the second half of reverse diffusion.

By diffusion index 500, the scheduled condition is already better across all three reconstruction metrics:

Guidance	**MSE ↓**	**PSNR ↑**	**LPIPS ↓**
Static m_r, index 500	0.1553	8.09 dB	0.9369
Scheduled m_r, index 500	**0.1143**	**9.42 dB**	**0.9218**

The separation is considerably larger at the end of the 1,000-step trajectory:

Guidance	**MSE ↓**	**PSNR ↑**	**LPIPS ↓**
Static m_r, final	0.1087	9.64 dB	0.9317
Scheduled m_r, final	**0.0548**	**12.61 dB**	**0.8744**

Scheduled guidance therefore ends with approximately **half the MSE** of fixed guidance and nearly **3 dB higher PSNR**, alongside a lower LPIPS score.

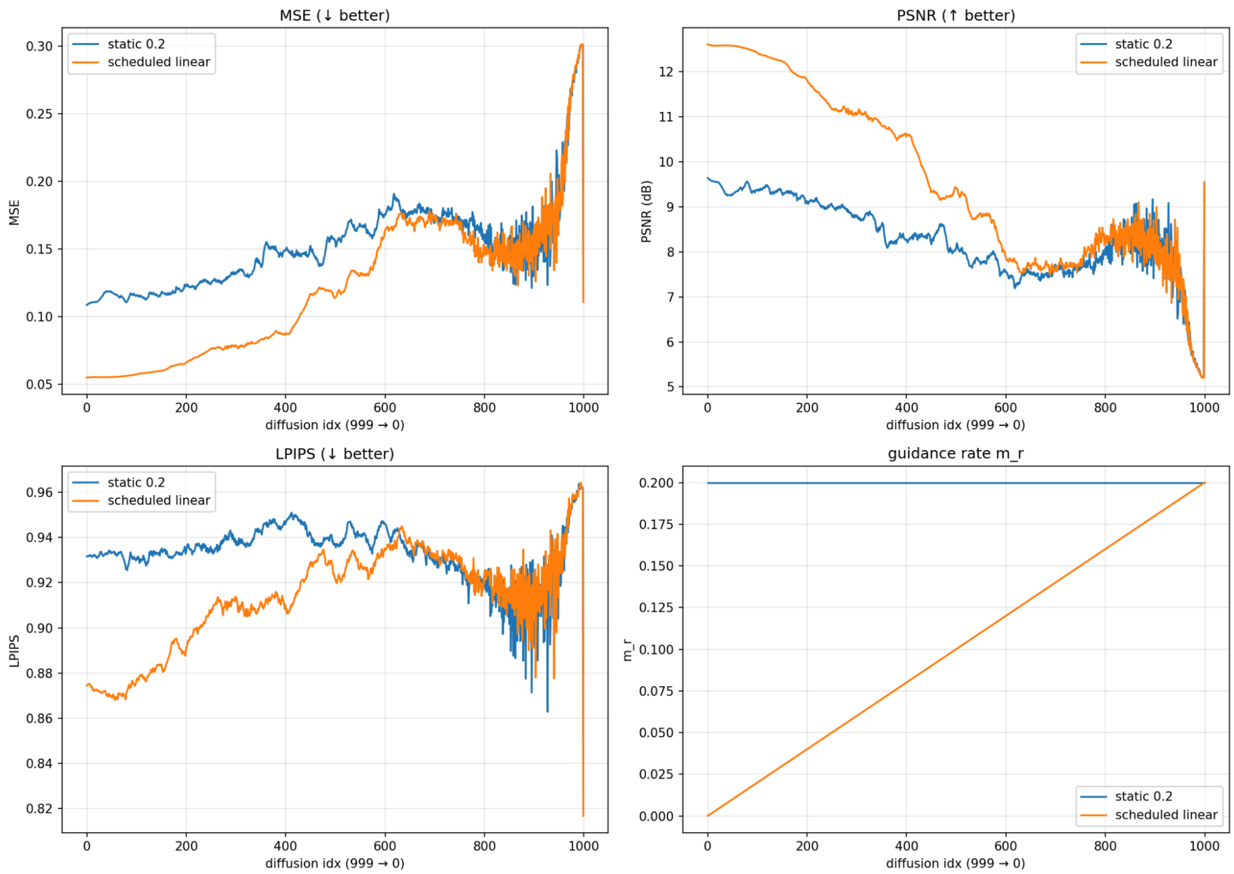

Figure: Trajectory of image-space metrics across the 1000-step GGSS-R reconstruction using fixed vs scheduled m_r.

The full metric trajectories show that the final result is part of a consistent trend rather than an isolated improvement. The two runs remain comparatively close during early denoising, when their guidance rates are still similar. As the scheduled m_r decreases, a clear gap develops between the curves. Scheduled guidance achieves lower MSE, higher PSNR, and lower LPIPS through the later reconstruction stages.

Figure: Image generated at the 500th timestep by ggss-r under static vs scheduled m_r.

The image trajectories support the numerical results. Both runs progressively recover facial structure, but the fixed-guidance samples become visibly more susceptible to the noisy gradient. In the 10×2 contact sheet, the scheduled reconstruction remains cleaner as reverse diffusion proceeds. The difference is already visible around the halfway point and is stronger in the final reconstruction.

The scheduled condition also achieves better best-observed MSE and PSNR.

The results indicate that a guidance strength useful at the beginning of noisy-gradient reconstruction can become too influential during later refinement. With fixed m_r = 0.2, the perturbed leaked gradient continues to exert the same guidance strength throughout the trajectory. Linear decay gradually transfers more control back to the diffusion prior, reducing the influence of the noise-contaminated gradient as reconstruction progresses.

For the tested **Gaussian noise variance 0.01**, scheduling m_r from 0.2 to 0 gives a clear improvement over the paper's fixed m_r = 0.2. The experiment therefore identifies **guidance scheduling as a useful extension to GGSS-R under gradient perturbation**.